

Detector Acceptance Study

Chatura Kuruppu

September 7, 2026

- Files Used & Event Selection
- Invariant Mass Studies
- x_F Studies (Yields & Acceptances)
- p_T Studies (Yields & Acceptances)
- Track Kinematics (dp_x, dp_y)
- Binned Slice Studies (x_F and p_T bins)

Files Used & Kinematic Bins

Files Used:

- LH2 and LD2 Drell-Yan Monte Carlo (M027_S001)
- **Thrown:** `mc_drellyan*_4pi_pTxFweight_v2.root`
- **Accepted:** `mc_drellyan*_clean_occ_pTxFweight_v2.root`

Acceptance Calculation:

$$\text{Acceptance} = \frac{\sum w_{\text{accepted}}}{\sum w_{\text{thrown}}} \quad \text{where } w = \text{ReWeight}$$

Kinematic Bin Boundaries:

- **Mass [GeV]:** [4.2, 4.5, 4.8, 5.1, 5.4, 5.7, 6.0, 6.3, 6.6, 6.9, 7.5, 8.8]
- x_F : [-0.05, 0.0, 0.05, ..., 0.85]
- p_T [GeV/c]: [0.0, 0.32, 0.49, 0.63, 0.77, 0.95, 1.18, 1.8]

Event Selection Criteria (Chuck's Cuts)

Dimuon Kinematics:

- $4.2 < \text{Mass} < 8.8 \text{ GeV}$
- $-0.1 < x_F < 0.95$
- $0.05 < x_T \leq 0.58$
- $38 < dp_z < 116 \text{ GeV}$
- $dp_x^2 + dp_y^2 < 5.0 \text{ GeV}^2$

Target & Vertex:

- $-280 < dz < -5 \text{ cm}$
- $|dx| < 0.25 \text{ cm}$
- $|dy - 1.6| < 0.22 \text{ cm}$

Track Quality:

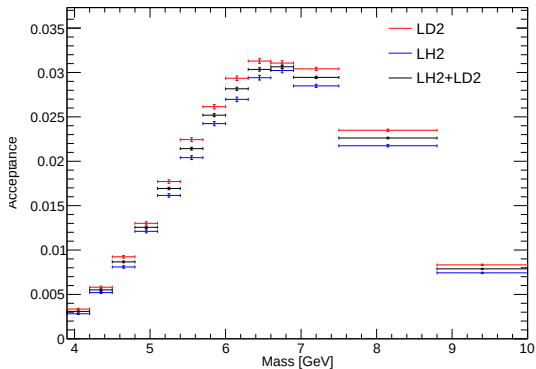
- $\chi_{\text{dimuon}}^2 < 18$
- Track separation $< 270 \text{ cm}$
- $p_{z,\text{st1}} \in [9, 75] \text{ GeV}$
- $n\text{Hits}_1 > 13, n\text{Hits}_2 > 13$

Generator-Level Fiducial Cuts (Thrown):

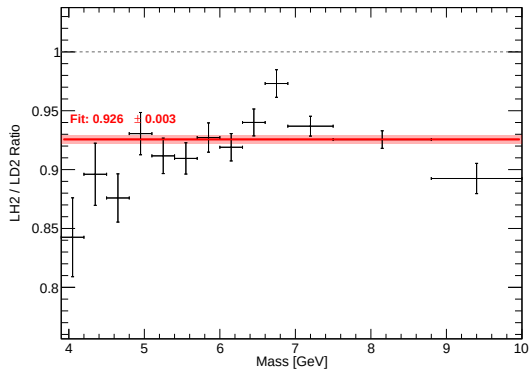
- $4.2 < \text{Mass} < 8.8 \text{ GeV}$
- $-0.1 < x_F < 0.95$
- $0.0 < p_T \leq 3.0 \text{ GeV}/c$

Overlay & Ratio: Mass (Integrated)

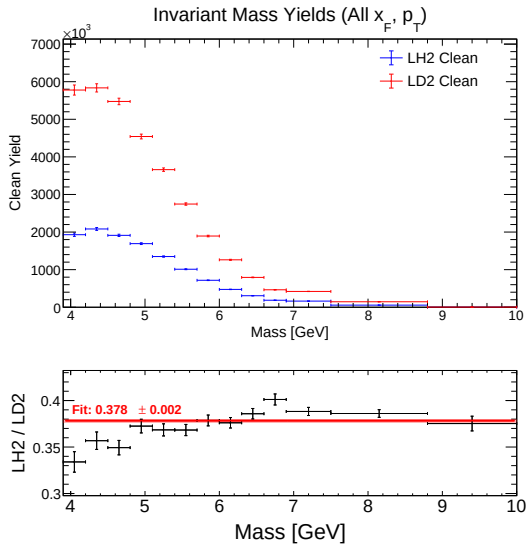
Integrated Acceptance vs mass



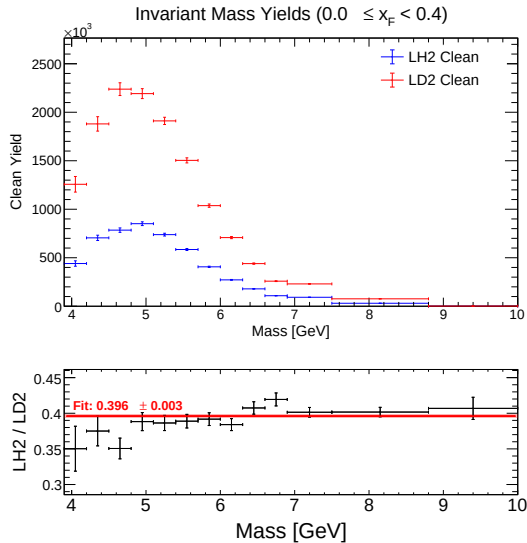
LH2/LD2 Integrated Acceptance vs mass



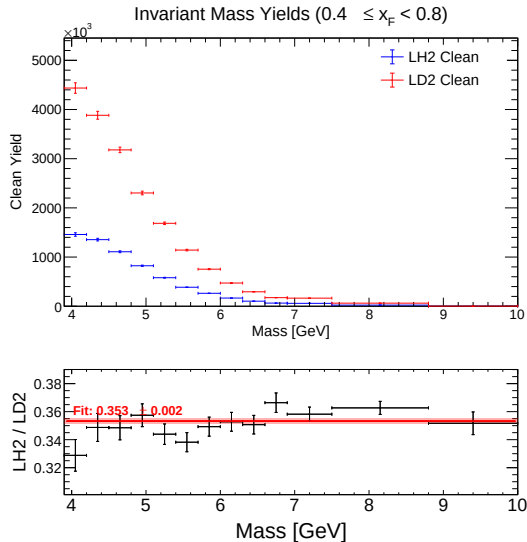
Invariant Mass Yields & Ratio (All x_F , p_T)



Invariant Mass Yields & Ratio ($0.0 < x_F \leq 0.4$)

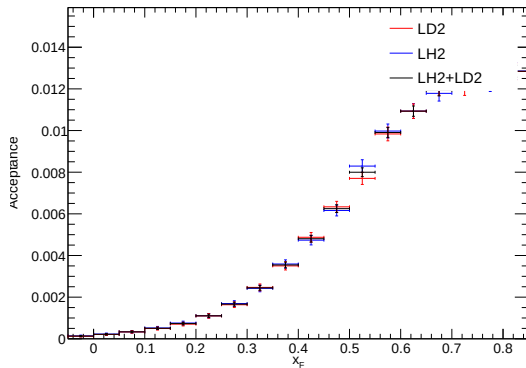


Invariant Mass Yields & Ratio ($0.4 < x_F \leq 0.8$)

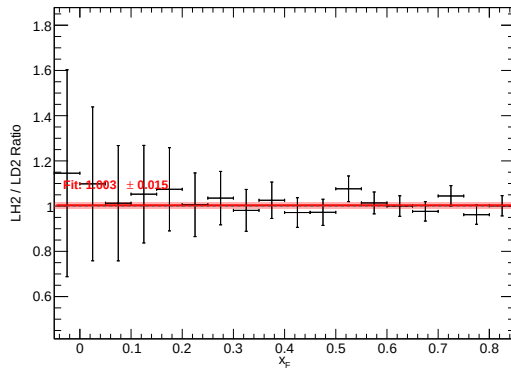


Overlay & Ratio: x_F (Integrated)

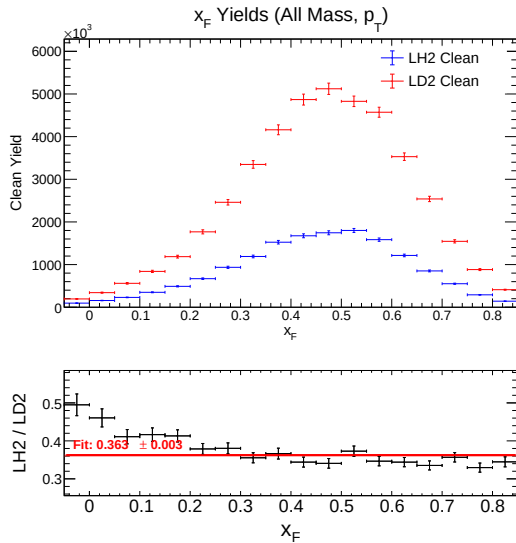
Integrated Acceptance vs x_F



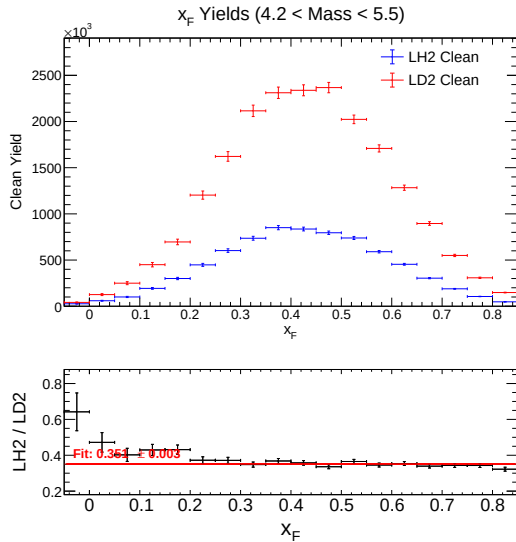
LH2/LD2 Integrated Acceptance vs x_F



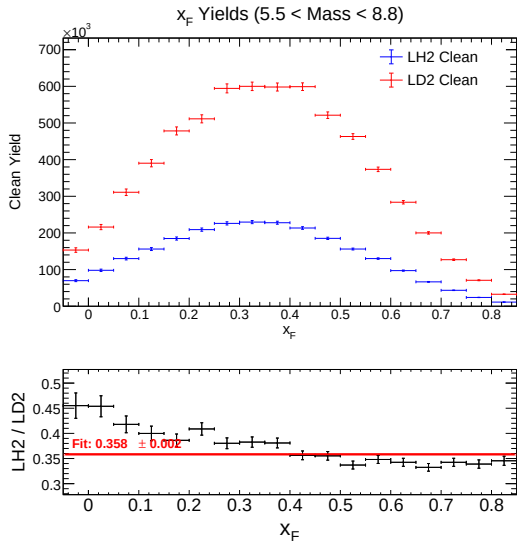
x_F Yields & Ratio (All Mass, p_T)



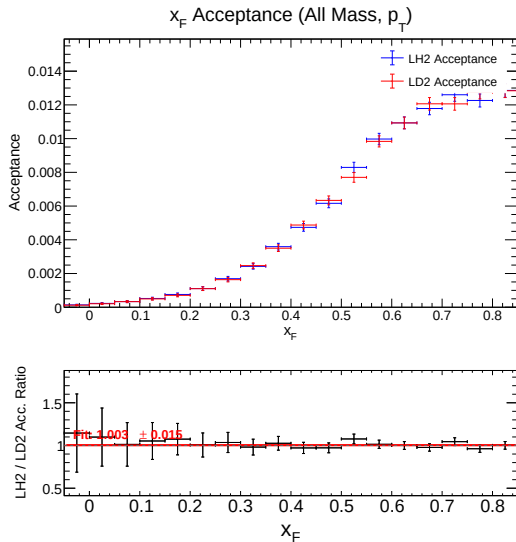
x_F Yields & Ratio ($4.2 < \text{Mass} \leq 5.5$)



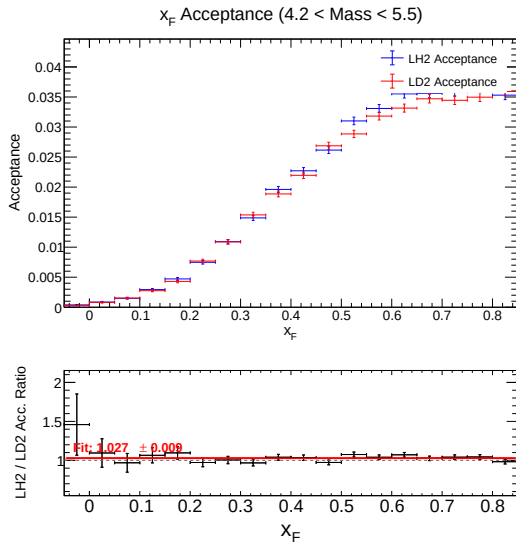
x_F Yields & Ratio ($5.5 < \text{Mass} \leq 8.8$)



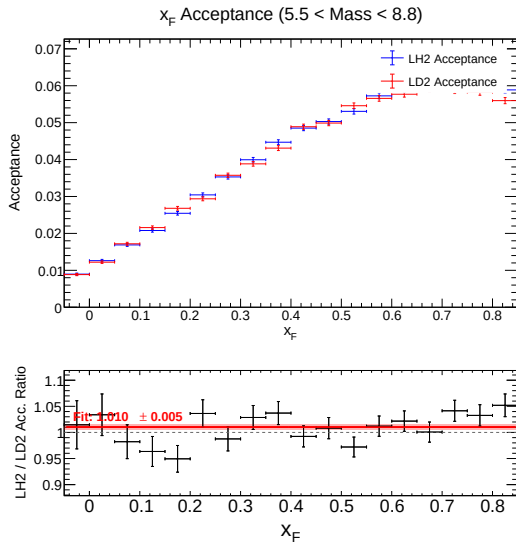
x_F Acceptance & Ratio (All Mass, p_T)



x_F Acceptance & Ratio ($4.2 < \text{Mass} < 5.5$)

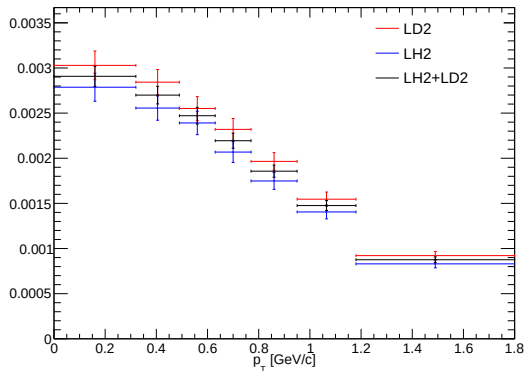


x_F Acceptance & Ratio ($5.5 < \text{Mass} < 8.8$)

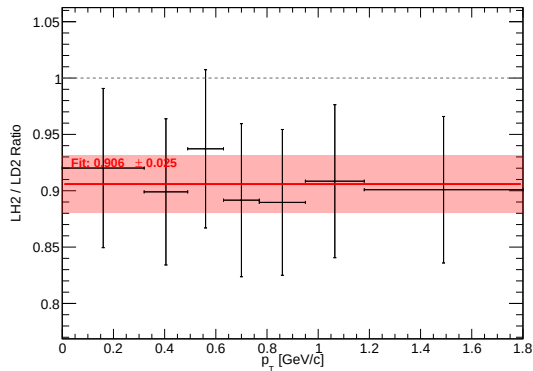


Overlay & Ratio: p_T (Integrated)

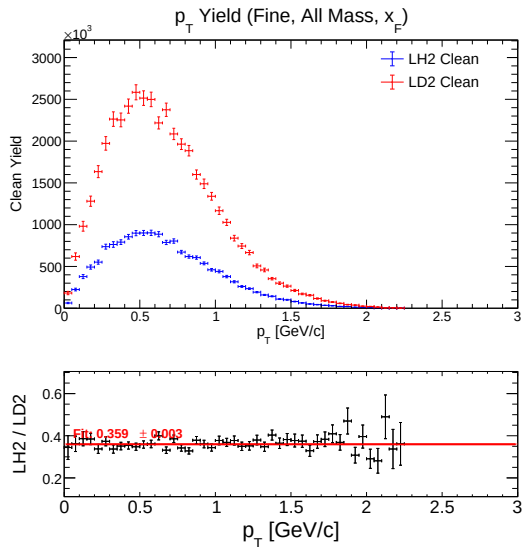
Integrated Acceptance vs p_T



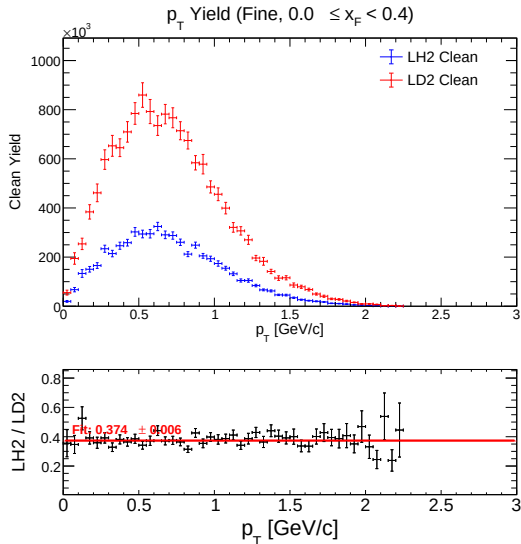
LH2/LD2 Integrated Acceptance vs p_T



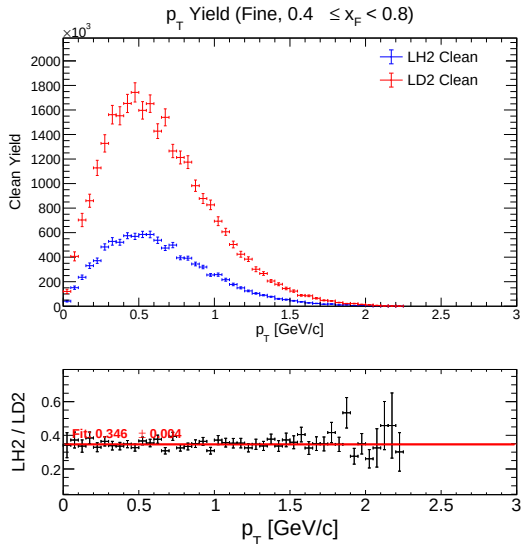
p_T Yields & Ratio (Fine Bins, All Mass, x_F)



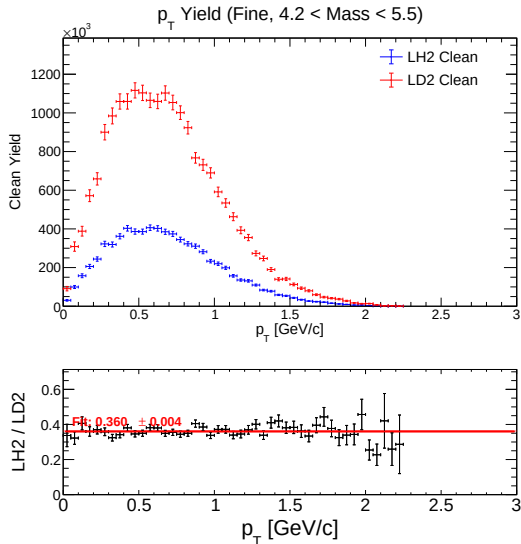
p_T Yields & Ratio (Fine Bins, $0.0 < x_F < 0.4$)



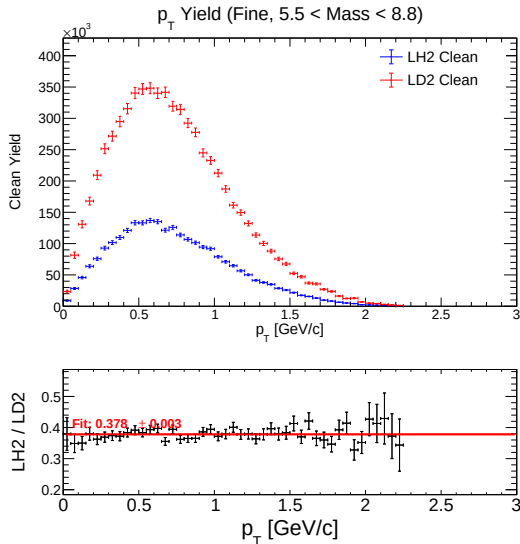
p_T Yields & Ratio (Fine Bins, $0.4 < x_F < 0.8$)



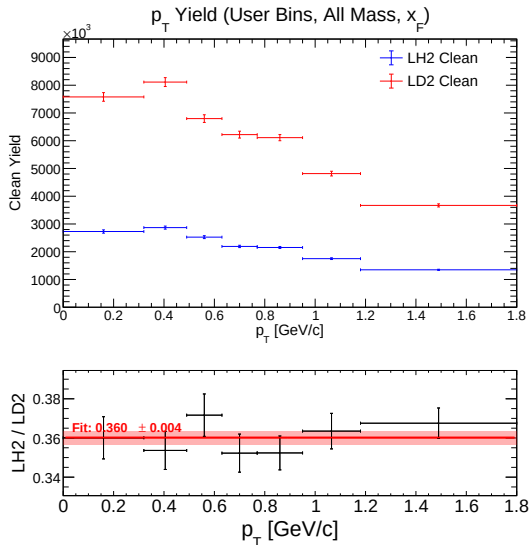
p_T Yields & Ratio (Fine Bins, $4.2 < \text{Mass} < 5.5$)



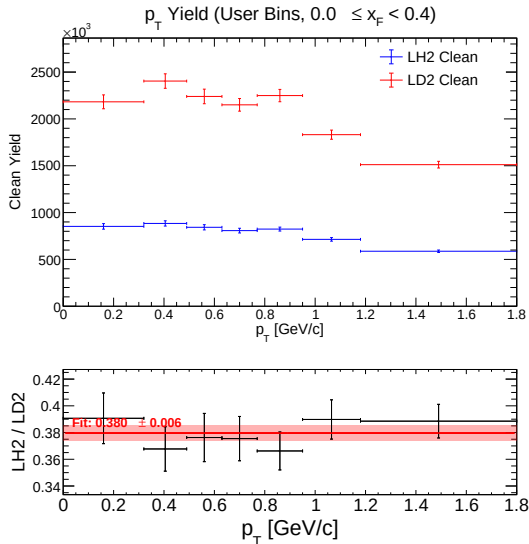
p_T Yields & Ratio (Fine Bins, $5.5 < \text{Mass} < 8.8$)



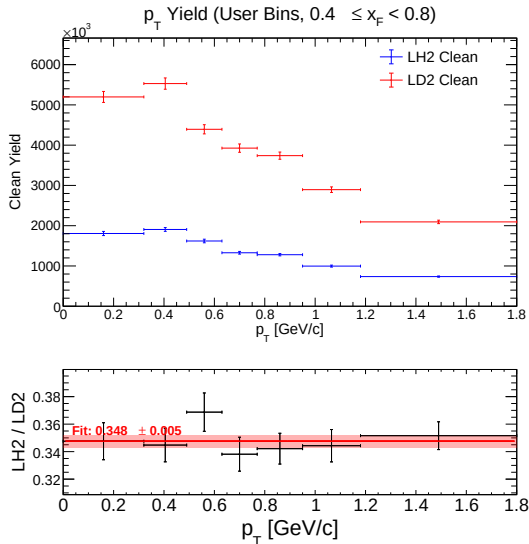
p_T Yields & Ratio (User Bins, All Mass, x_F)



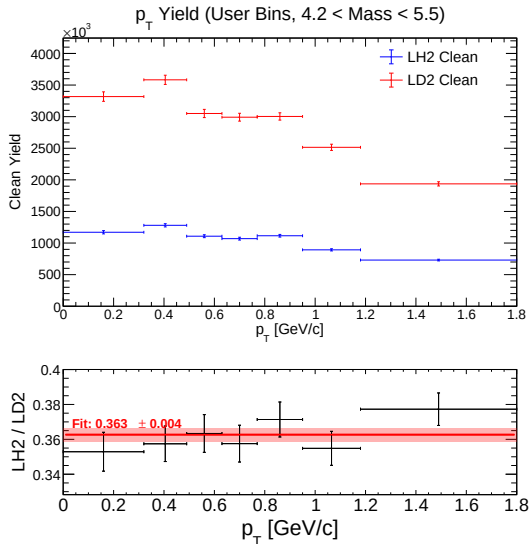
p_T Yields & Ratio (User Bins, $0.0 < x_F < 0.4$)



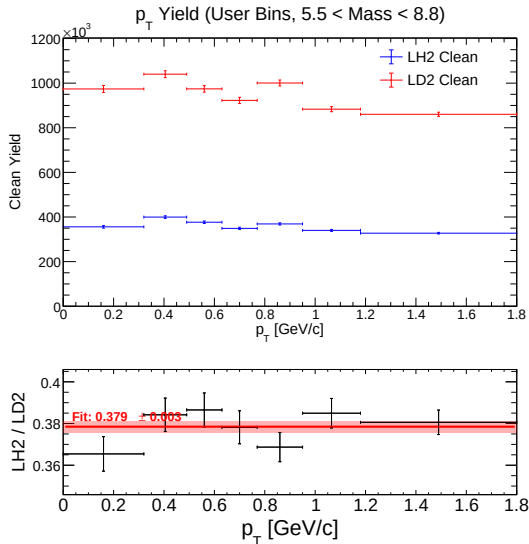
p_T Yields & Ratio (User Bins, $0.4 < x_F < 0.8$)



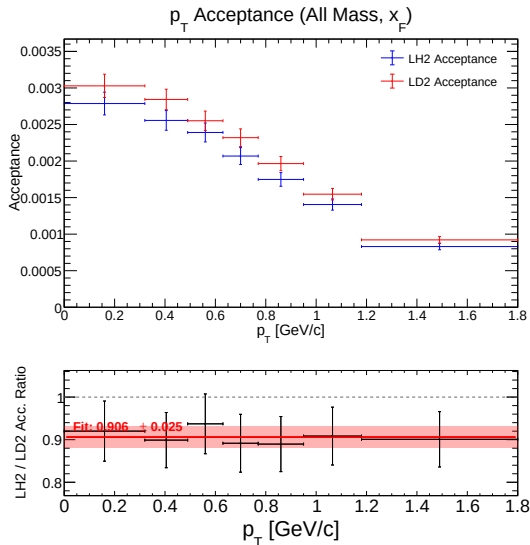
p_T Yields & Ratio (User Bins, $4.2 < \text{Mass} < 5.5$)



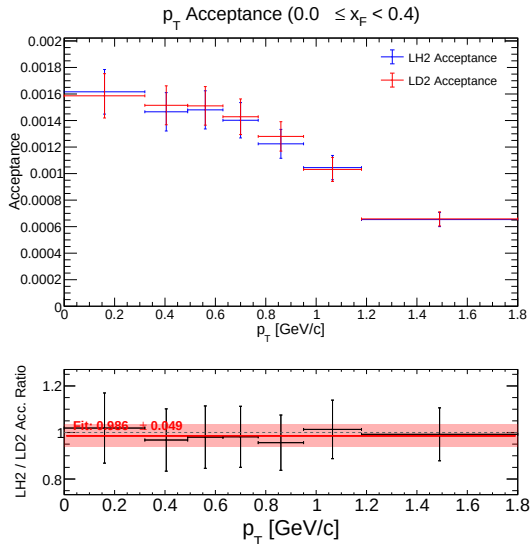
p_T Yields & Ratio (User Bins, $5.5 < \text{Mass} < 8.8$)



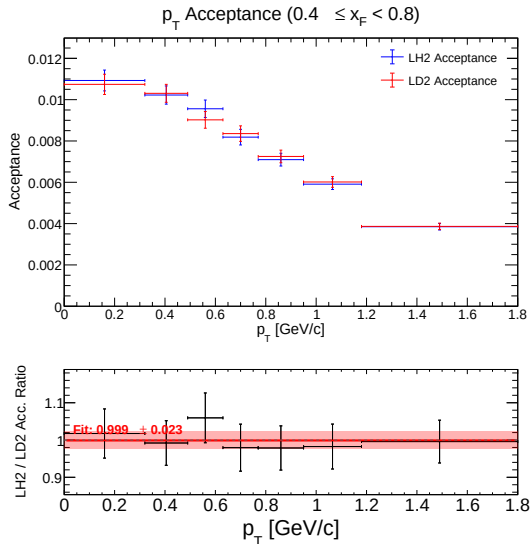
p_T Acceptance & Ratio (User Bins, All Mass, x_F)



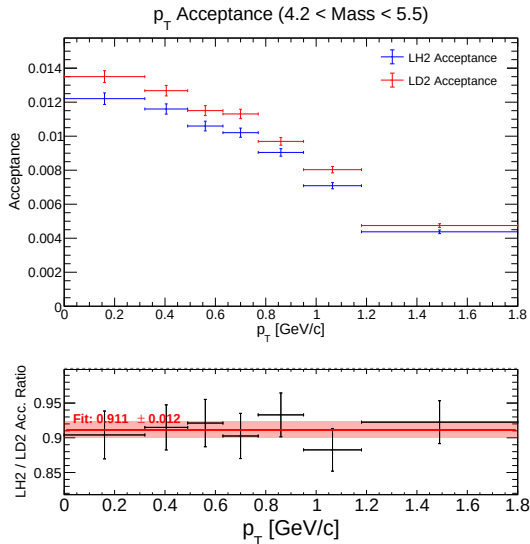
p_T Acceptance & Ratio (User Bins, $0.0 < x_F < 0.4$)



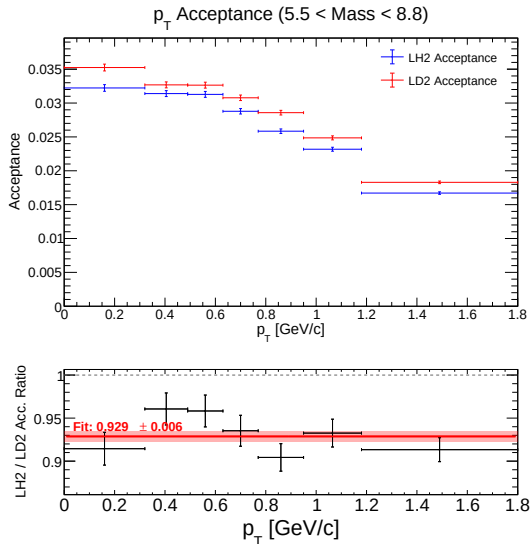
p_T Acceptance & Ratio (User Bins, $0.4 < x_F < 0.8$)



p_T Acceptance & Ratio (User Bins, $4.2 < \text{Mass} < 5.5$)

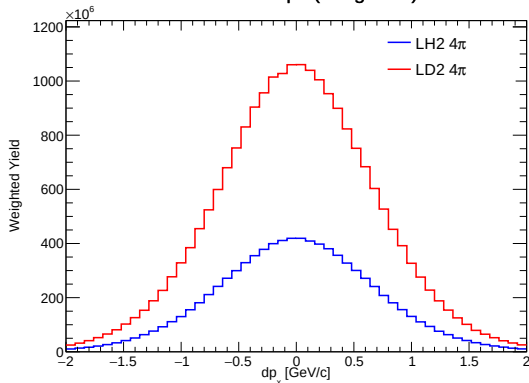


p_T Acceptance & Ratio (User Bins, $5.5 < \text{Mass} < 8.8$)

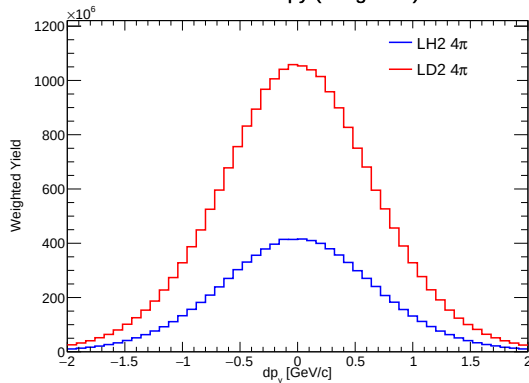


Kinematic Yields (4π Thrown): Integrated

4π Yield of dp_x (Integrated)

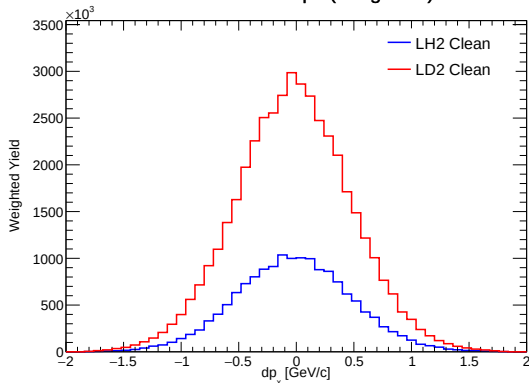


4π Yield of dp_y (Integrated)

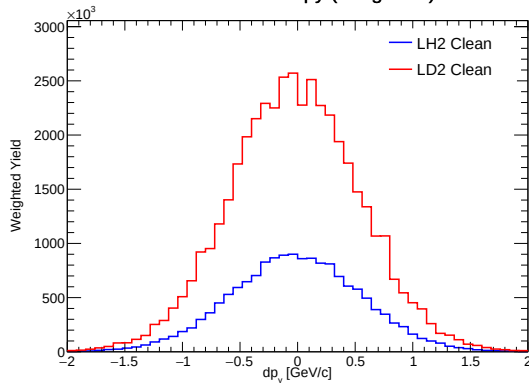


Kinematic Yields (Clean Accepted): Integrated

Clean Yield of dp_x (Integrated)

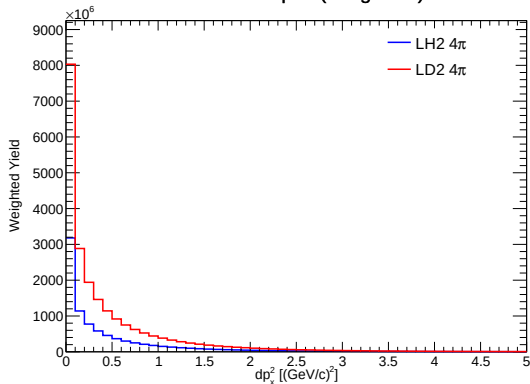


Clean Yield of dp_y (Integrated)

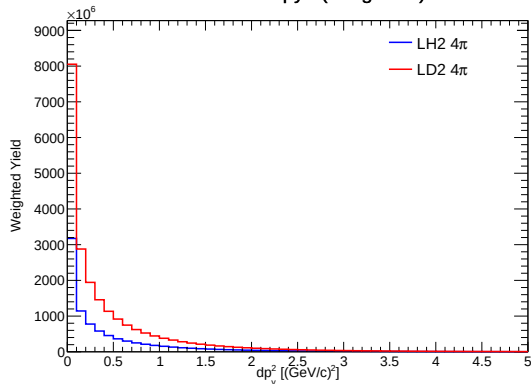


Kinematic Yields Squared (4π Thrown): Integrated

4π Yield of dp_x^2 (Integrated)

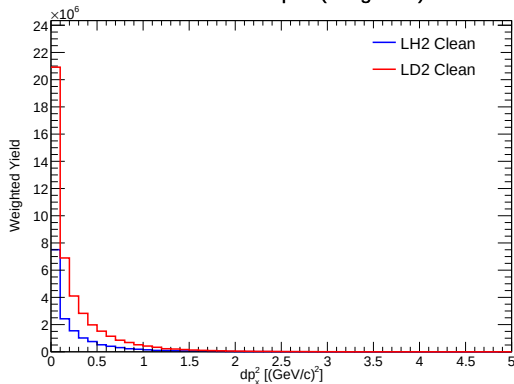


4π Yield of dp_y^2 (Integrated)

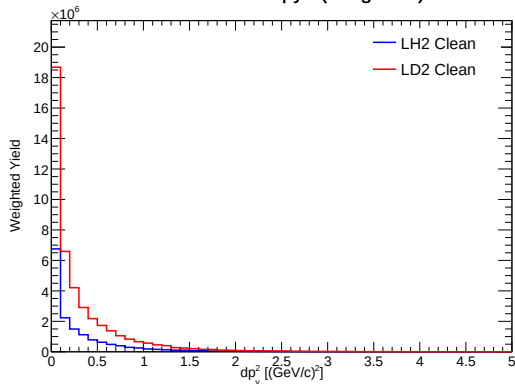


Kinematic Yields Squared (Clean Accepted): Integrated

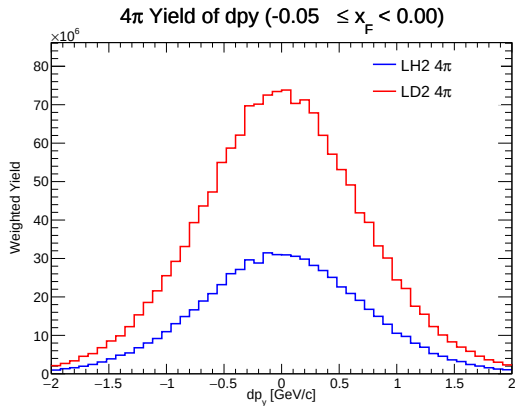
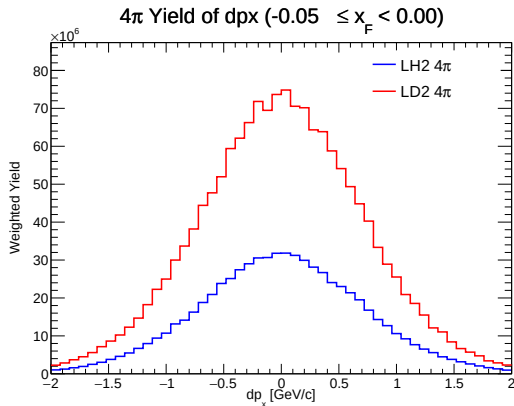
Clean Yield of dp_x^2 (Integrated)



Clean Yield of dp_y^2 (Integrated)

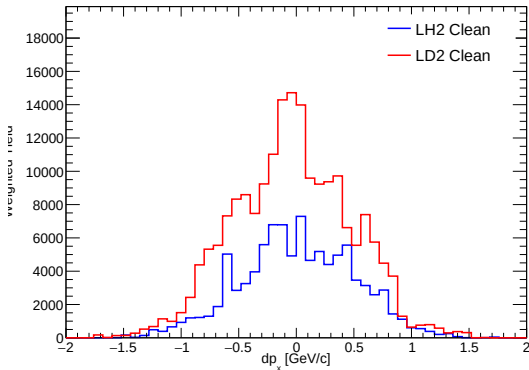


Kinematic Yields (4π Thrown): x_F Bin 0

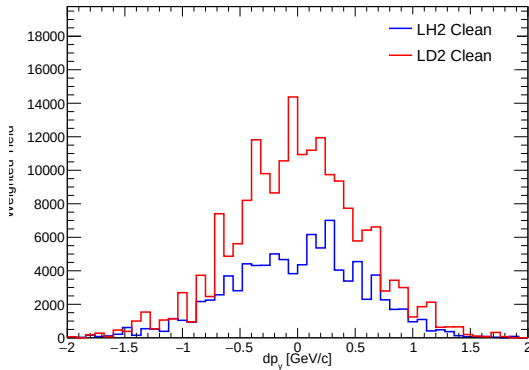


Kinematic Yields (Clean Accepted): x_F Bin 0

Clean Yield of dp_x ($-0.05 \leq x_F < 0.00$)

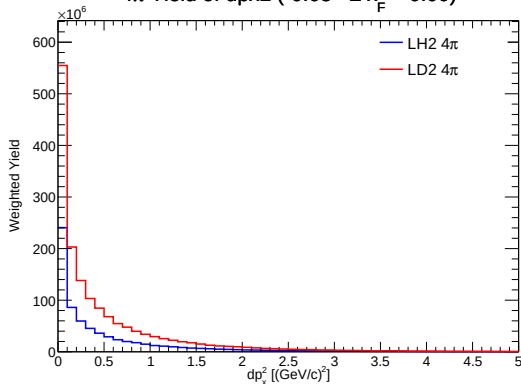


Clean Yield of dp_y ($-0.05 \leq x_F < 0.00$)

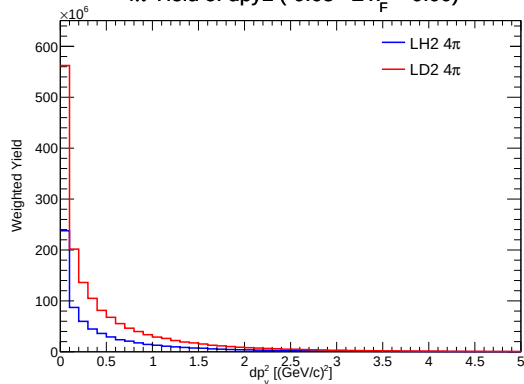


Kinematic Yields Squared (4π Thrown): x_F Bin 0

4π Yield of dp_x^2 ($-0.05 \leq x_F < 0.00$)

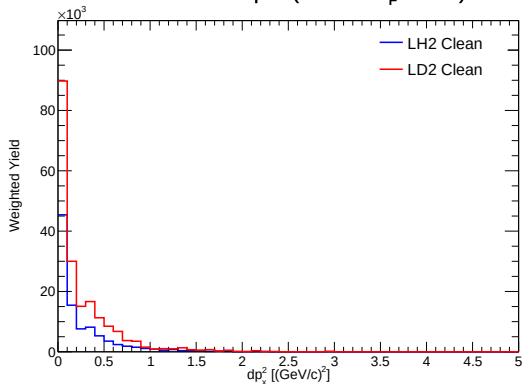


4π Yield of dp_y^2 ($-0.05 \leq x_F < 0.00$)

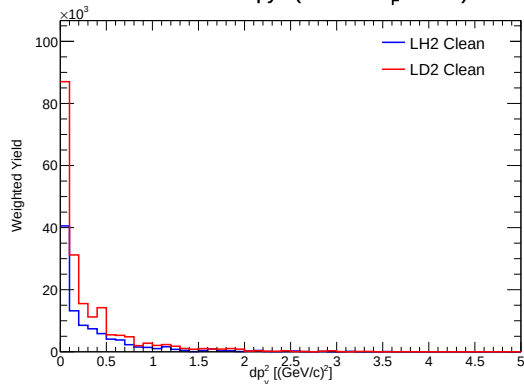


Kinematic Yields Squared (Clean Accepted): x_F Bin 0

Clean Yield of dp_x^2 ($-0.05 \leq x_F < 0.00$)

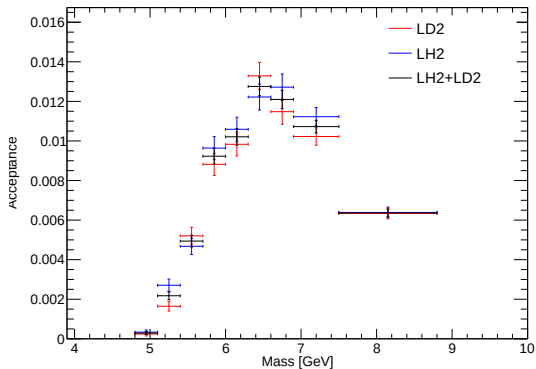


Clean Yield of dp_y^2 ($-0.05 \leq x_F < 0.00$)

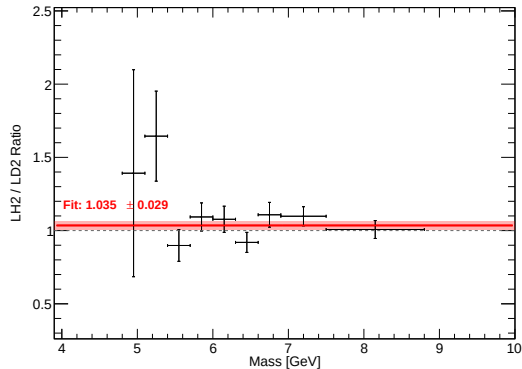


Overlay & Ratio (Mass sliced by x_F): x_F Bin 0

$-0.05 \leq x_F < 0.00$

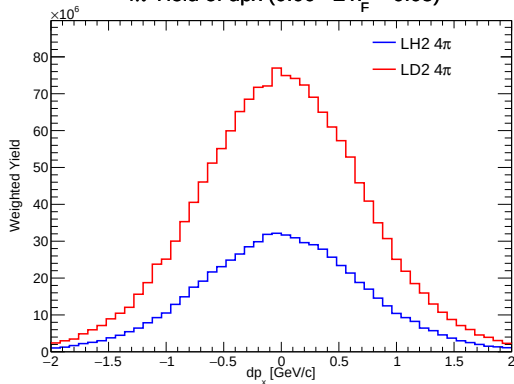


LH2/LD2 $-0.05 \leq x_F < 0.00$

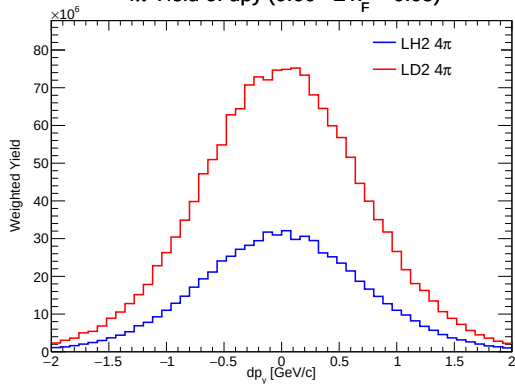


Kinematic Yields (4π Thrown): x_F Bin 1

4π Yield of dp_x ($0.00 \leq x_F < 0.05$)

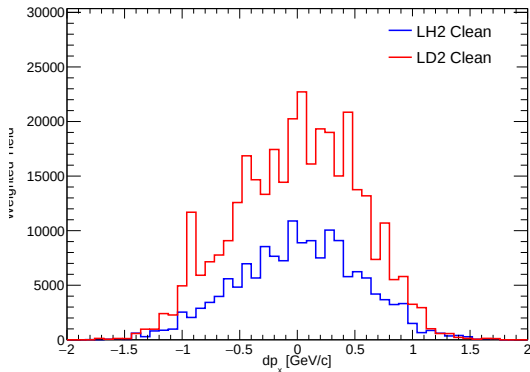


4π Yield of dp_y ($0.00 \leq x_F < 0.05$)

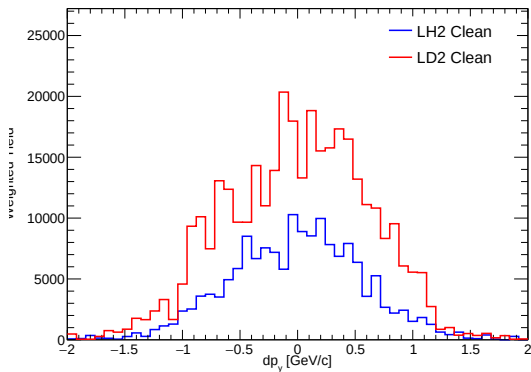


Kinematic Yields (Clean Accepted): x_F Bin 1

Clean Yield of dp_x ($0.00 \leq x_F < 0.05$)

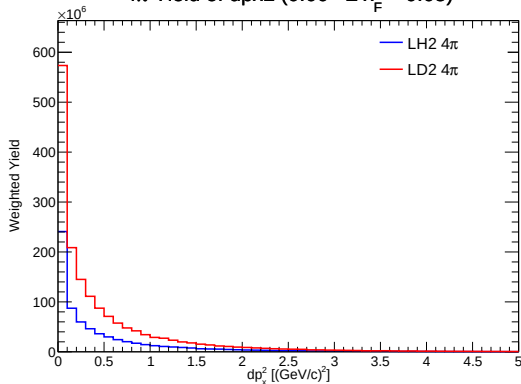


Clean Yield of dp_y ($0.00 \leq x_F < 0.05$)

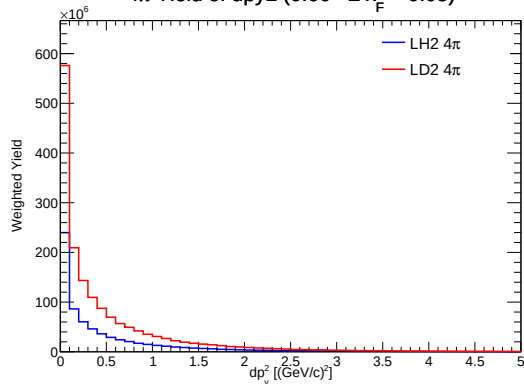


Kinematic Yields Squared (4π Thrown): x_F Bin 1

4 π Yield of dp_x^2 ($0.00 \leq x_F < 0.05$)

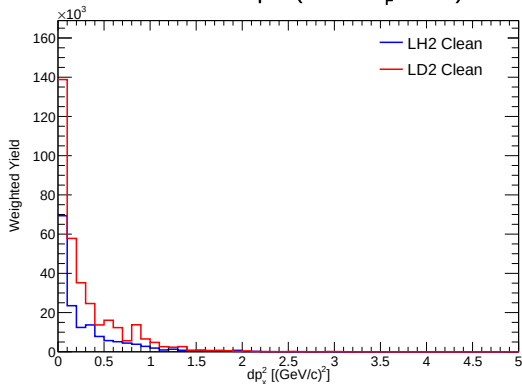


4 π Yield of dp_y^2 ($0.00 \leq x_F < 0.05$)

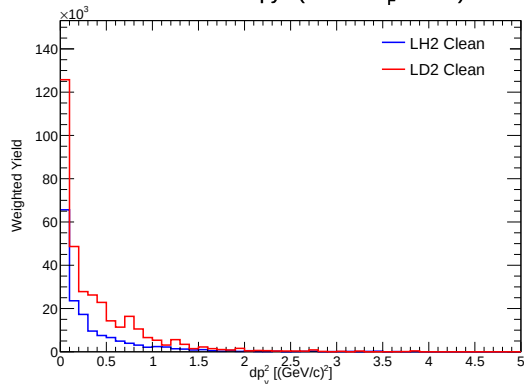


Kinematic Yields Squared (Clean Accepted): x_F Bin 1

Clean Yield of dp_x^2 ($0.00 \leq x_F < 0.05$)

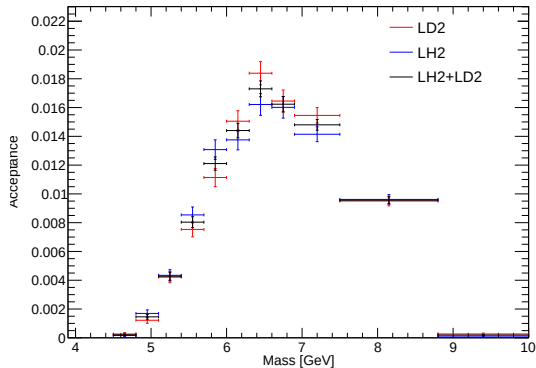


Clean Yield of dp_y^2 ($0.00 \leq x_F < 0.05$)

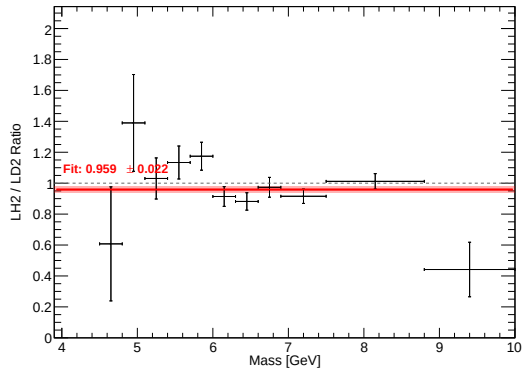


Overlay & Ratio (Mass sliced by x_F): x_F Bin 1

$0.00 \leq x_F < 0.05$

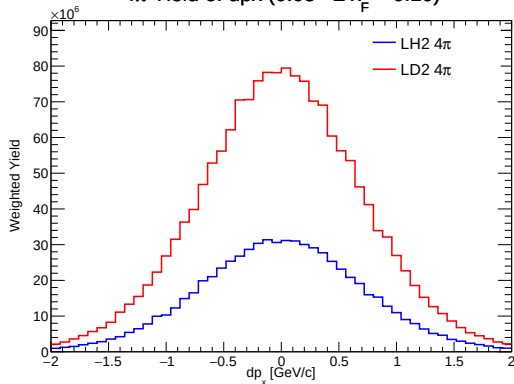


LH2/LD2 $0.00 \leq x_F < 0.05$

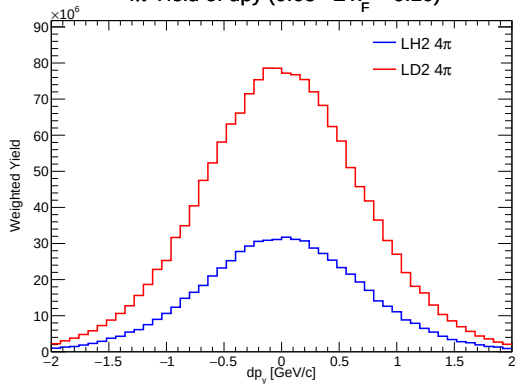


Kinematic Yields (4π Thrown): x_F Bin 2

4π Yield of dp_x ($0.05 \leq x_F < 0.10$)

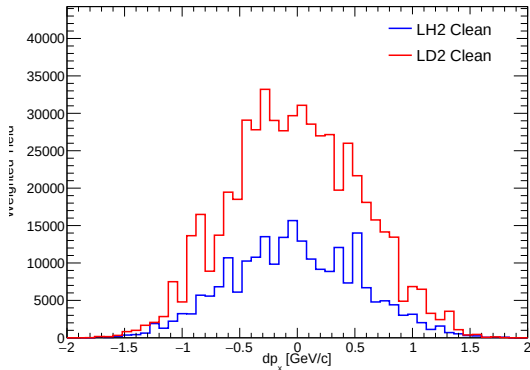


4π Yield of dp_y ($0.05 \leq x_F < 0.10$)

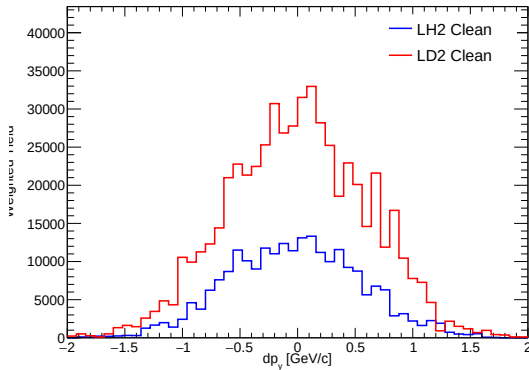


Kinematic Yields (Clean Accepted): x_F Bin 2

Clean Yield of dp_x ($0.05 \leq x_F < 0.10$)

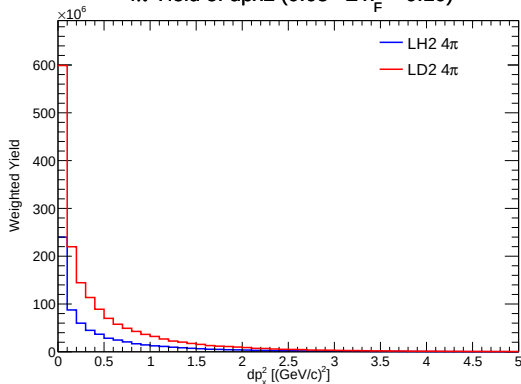


Clean Yield of dp_y ($0.05 \leq x_F < 0.10$)

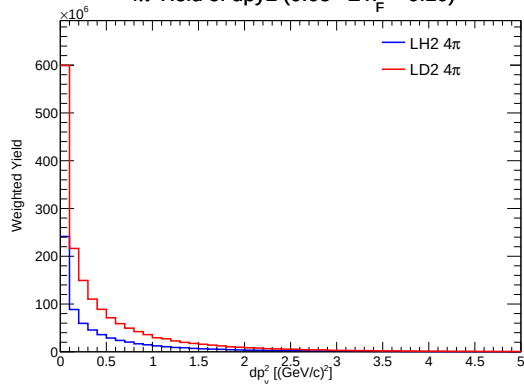


Kinematic Yields Squared (4π Thrown): x_F Bin 2

4 π Yield of dp_x^2 ($0.05 \leq x_F < 0.10$)

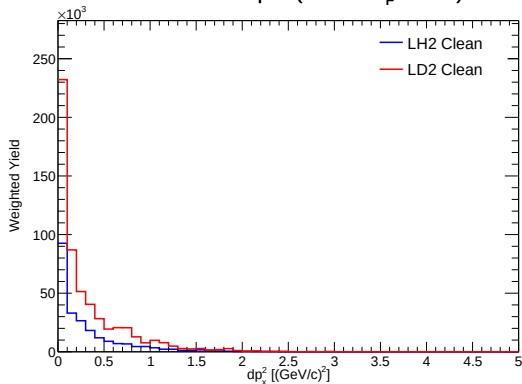


4 π Yield of dp_y^2 ($0.05 \leq x_F < 0.10$)

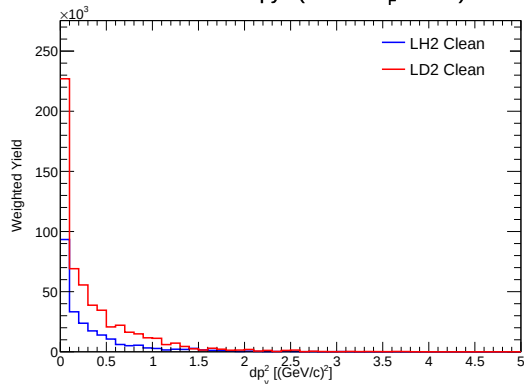


Kinematic Yields Squared (Clean Accepted): x_F Bin 2

Clean Yield of dp_x^2 ($0.05 \leq x_F < 0.10$)

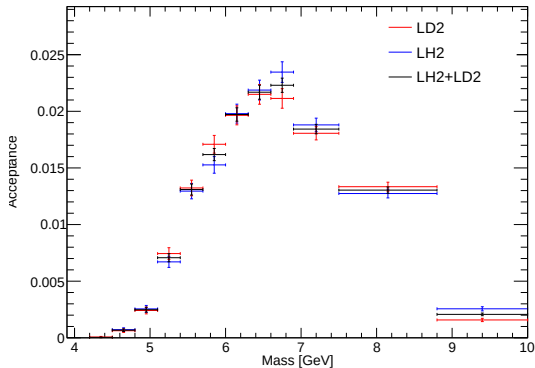


Clean Yield of dp_y^2 ($0.05 \leq x_F < 0.10$)

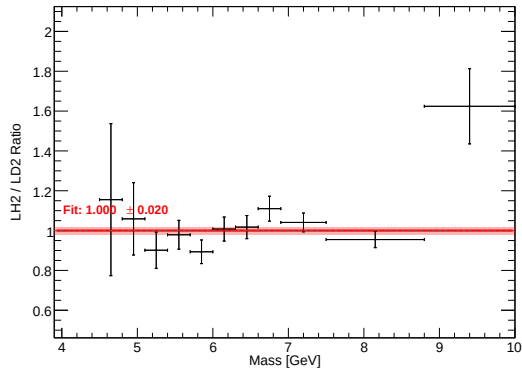


Overlay & Ratio (Mass sliced by x_F): x_F Bin 2

$0.05 \leq x_F < 0.10$

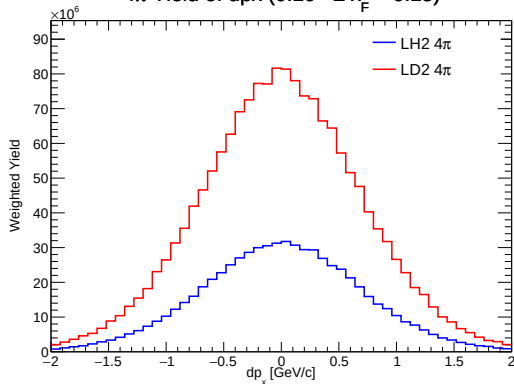


LH2/LD2 $0.05 \leq x_F < 0.10$

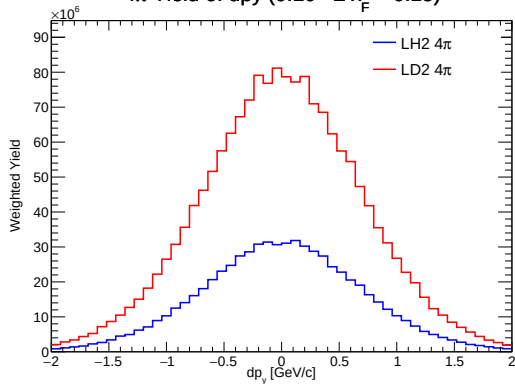


Kinematic Yields (4π Thrown): x_F Bin 3

4π Yield of dp_x ($0.10 \leq x_F < 0.15$)

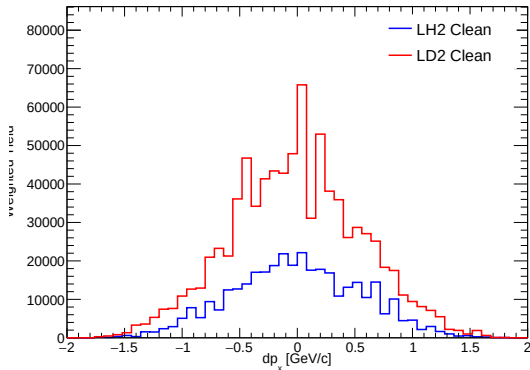


4π Yield of dp_y ($0.10 \leq x_F < 0.15$)

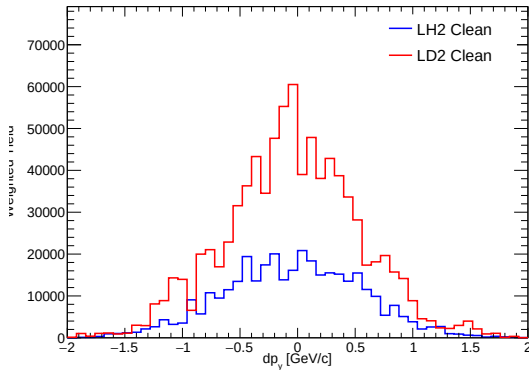


Kinematic Yields (Clean Accepted): x_F Bin 3

Clean Yield of dp_x ($0.10 \leq x_F < 0.15$)

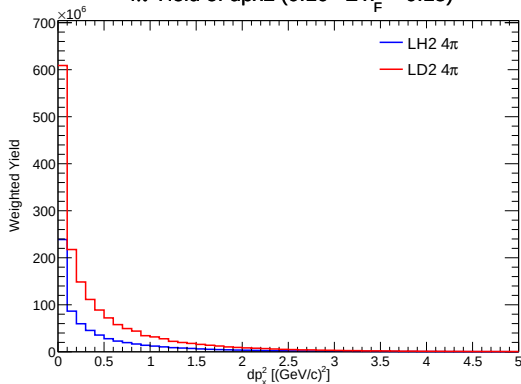


Clean Yield of dp_y ($0.10 \leq x_F < 0.15$)

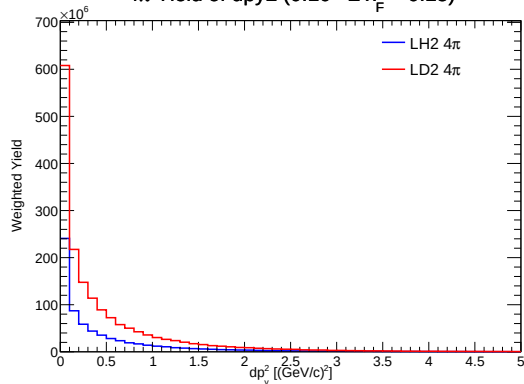


Kinematic Yields Squared (4π Thrown): x_F Bin 3

4 π Yield of dp_x^2 ($0.10 \leq x_F < 0.15$)

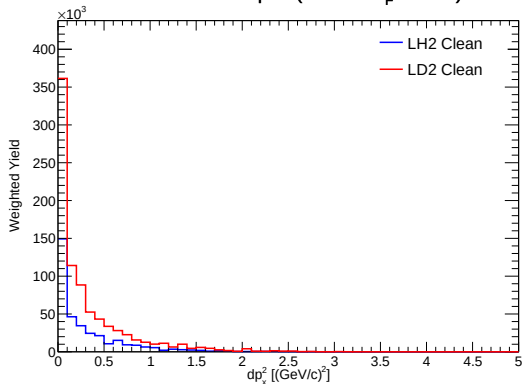


4 π Yield of dp_y^2 ($0.10 \leq x_F < 0.15$)

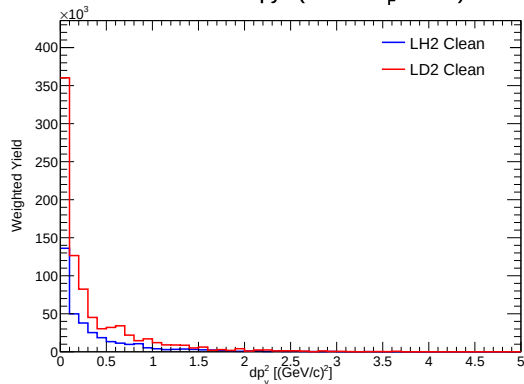


Kinematic Yields Squared (Clean Accepted): x_F Bin 3

Clean Yield of dp_x^2 ($0.10 \leq x_F < 0.15$)

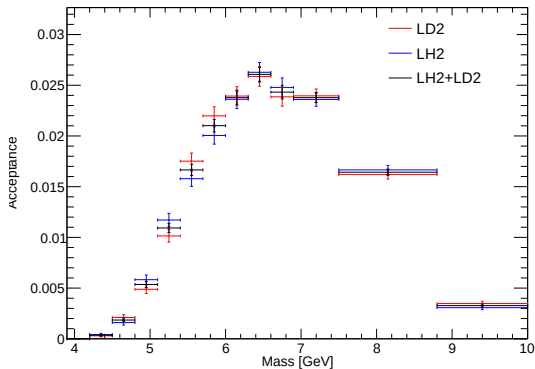


Clean Yield of dp_y^2 ($0.10 \leq x_F < 0.15$)

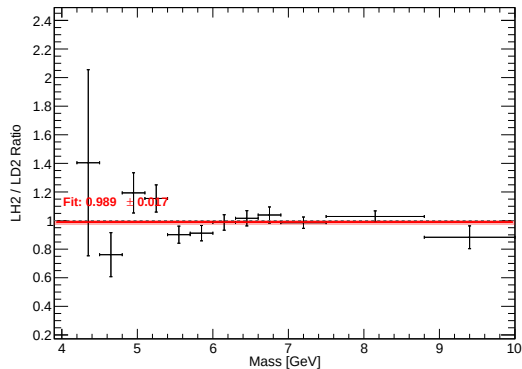


Overlay & Ratio (Mass sliced by x_F): x_F Bin 3

$0.10 \leq x_F < 0.15$

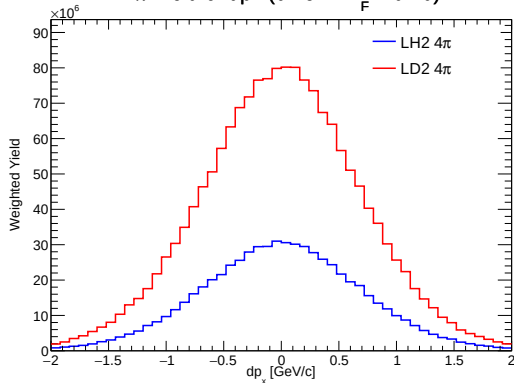


LH2/LD2 $0.10 \leq x_F < 0.15$

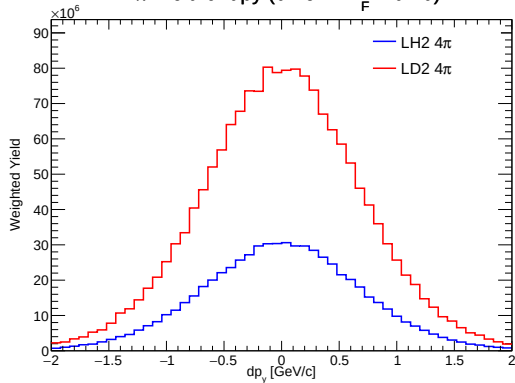


Kinematic Yields (4π Thrown): x_F Bin 4

4π Yield of dp_x ($0.15 \leq x_F < 0.20$)

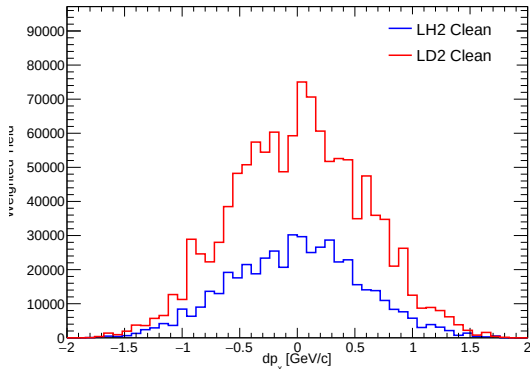


4π Yield of dp_y ($0.15 \leq x_F < 0.20$)

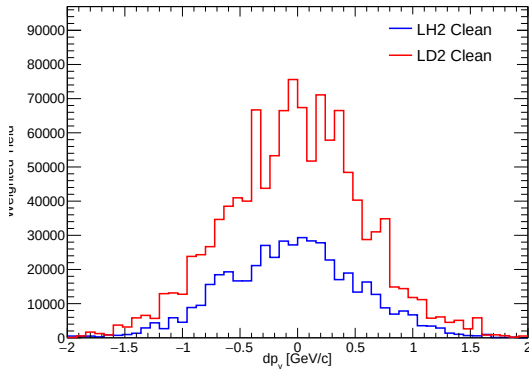


Kinematic Yields (Clean Accepted): x_F Bin 4

Clean Yield of dp_x ($0.15 \leq x_F < 0.20$)

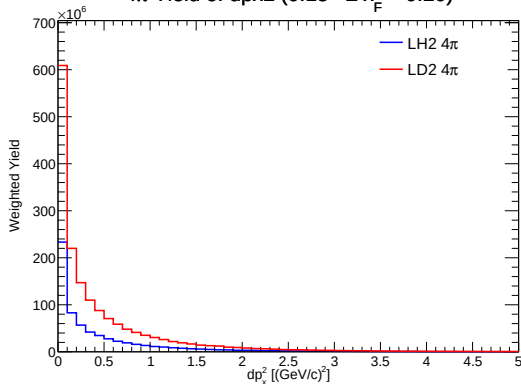


Clean Yield of dp_y ($0.15 \leq x_F < 0.20$)

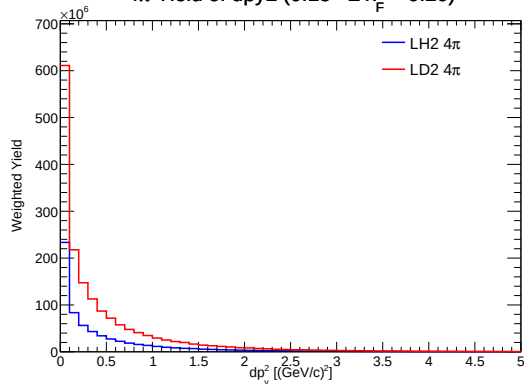


Kinematic Yields Squared (4π Thrown): x_F Bin 4

4 π Yield of dp_x^2 ($0.15 \leq x_F < 0.20$)

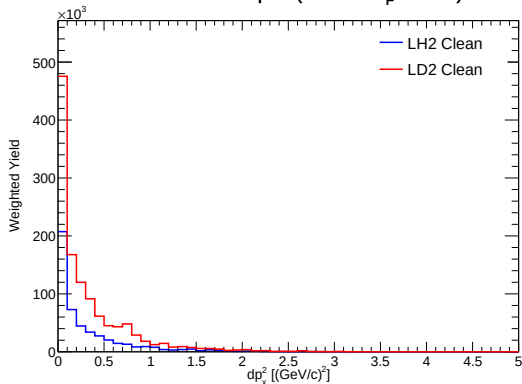


4 π Yield of dp_y^2 ($0.15 \leq x_F < 0.20$)

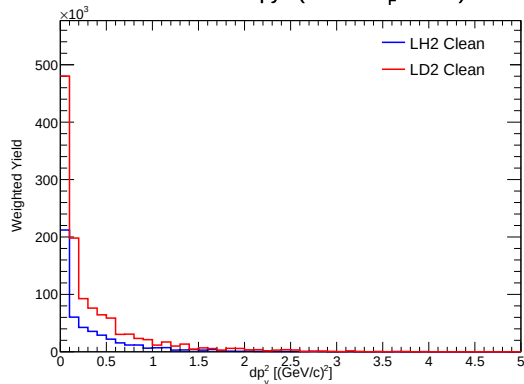


Kinematic Yields Squared (Clean Accepted): x_F Bin 4

Clean Yield of dp_x^2 ($0.15 \leq x_F < 0.20$)

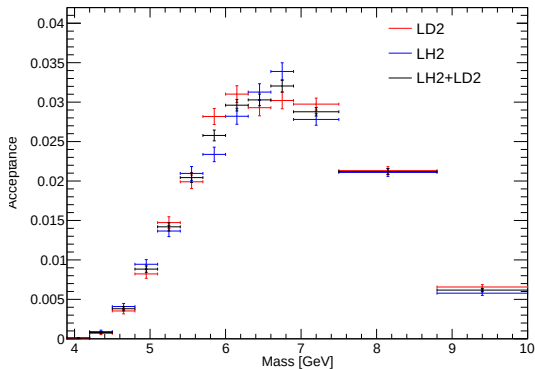


Clean Yield of dp_y^2 ($0.15 \leq x_F < 0.20$)

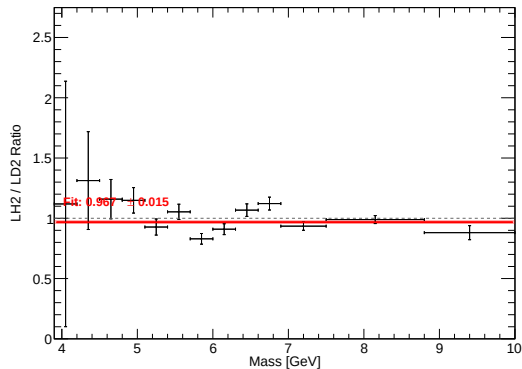


Overlay & Ratio (Mass sliced by x_F): x_F Bin 4

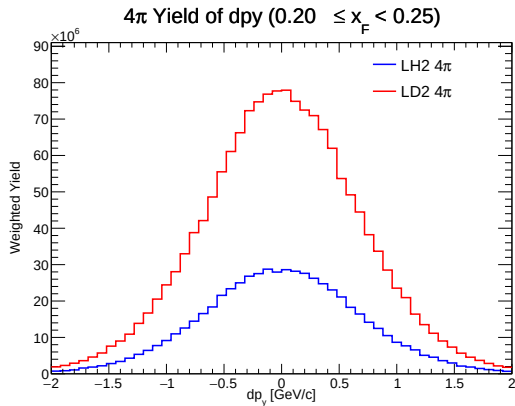
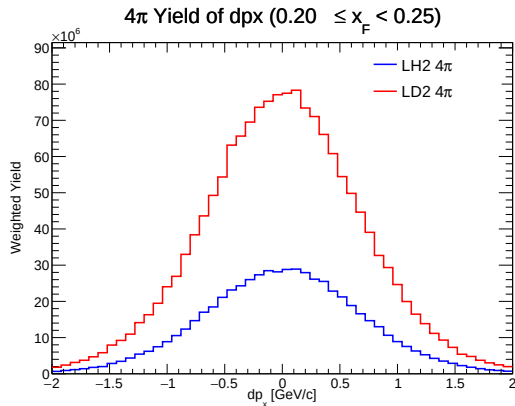
$0.15 \leq x_F < 0.20$



LH2/LD2 $0.15 \leq x_F < 0.20$

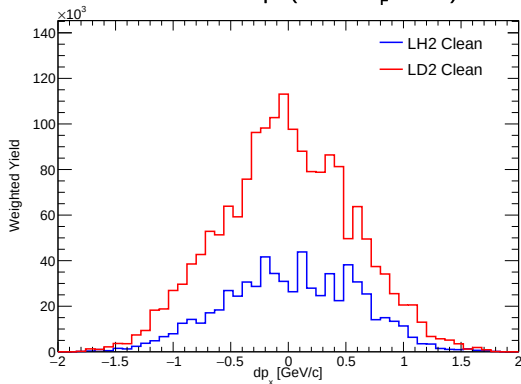


Kinematic Yields (4π Thrown): x_F Bin 5

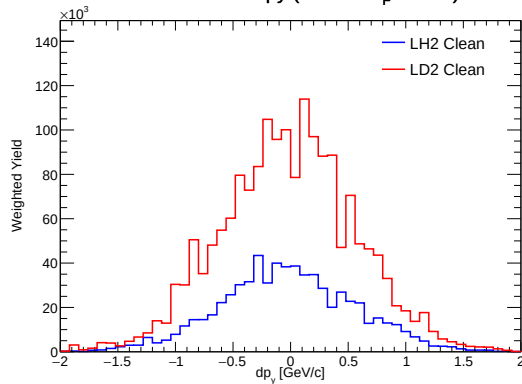


Kinematic Yields (Clean Accepted): x_F Bin 5

Clean Yield of dp_x ($0.20 \leq x_F < 0.25$)

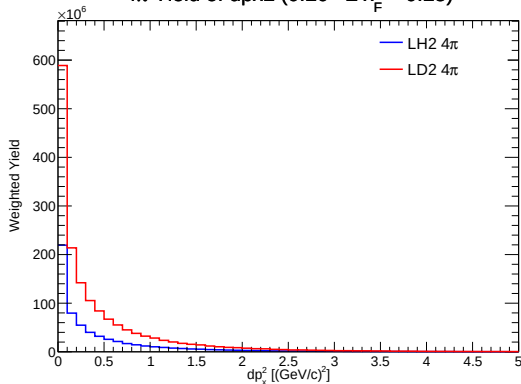


Clean Yield of dp_y ($0.20 \leq x_F < 0.25$)

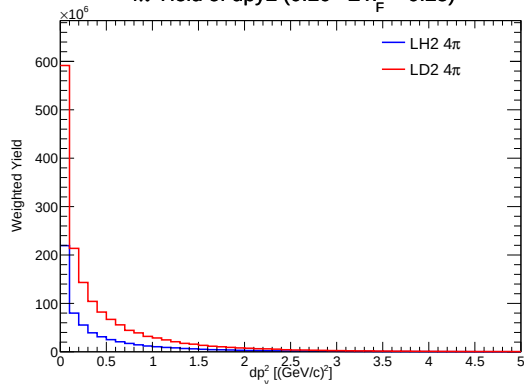


Kinematic Yields Squared (4π Thrown): x_F Bin 5

4 π Yield of dp_x^2 ($0.20 \leq x_F < 0.25$)

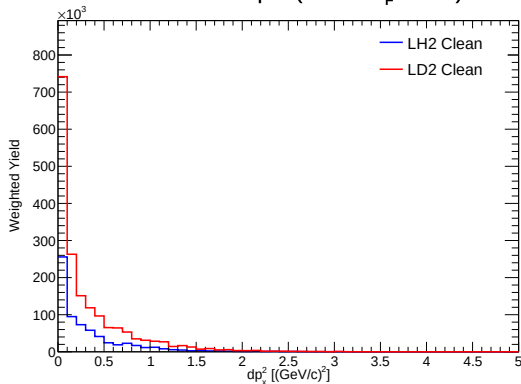


4 π Yield of dp_y^2 ($0.20 \leq x_F < 0.25$)

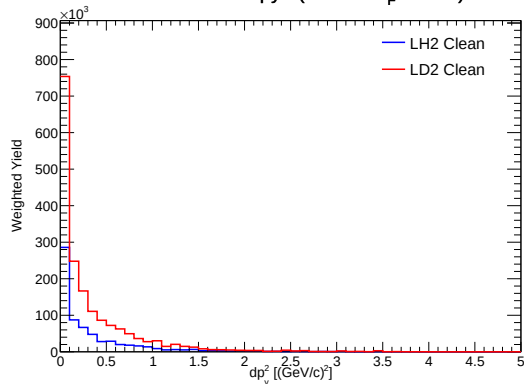


Kinematic Yields Squared (Clean Accepted): x_F Bin 5

Clean Yield of dp_x^2 ($0.20 \leq x_F < 0.25$)

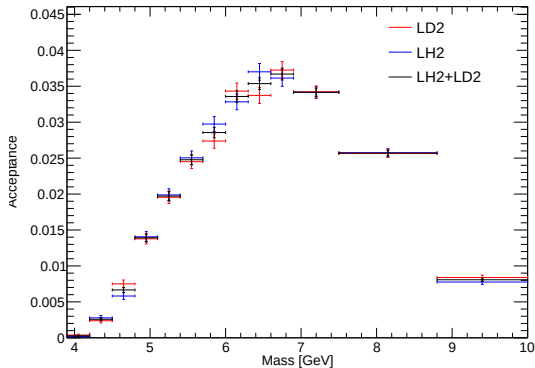


Clean Yield of dp_y^2 ($0.20 \leq x_F < 0.25$)

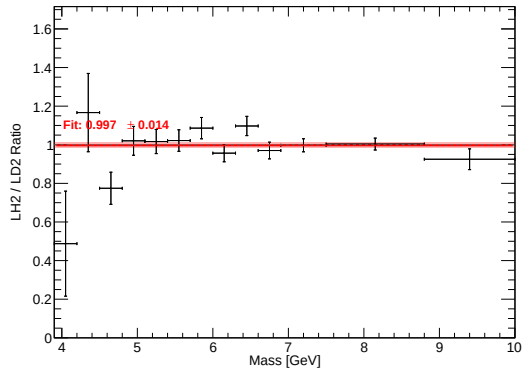


Overlay & Ratio (Mass sliced by x_F): x_F Bin 5

$0.20 \leq x_F < 0.25$

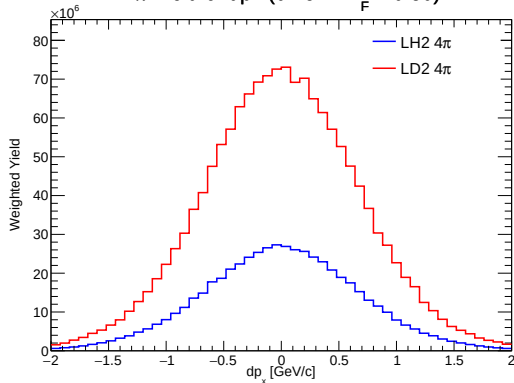


LH2/LD2 $0.20 \leq x_F < 0.25$

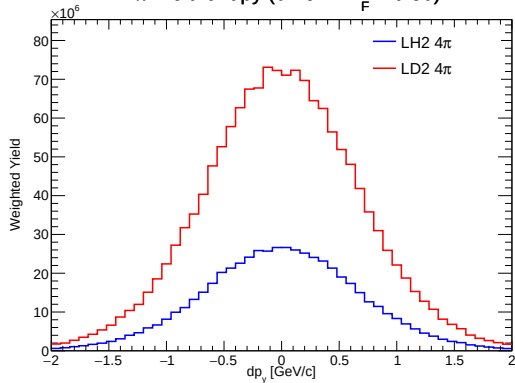


Kinematic Yields (4π Thrown): x_F Bin 6

4π Yield of dp_x ($0.25 \leq x_F < 0.30$)

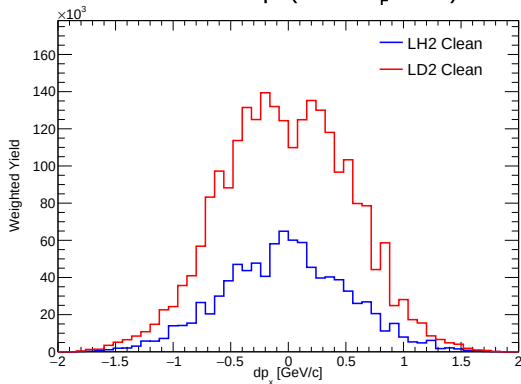


4π Yield of dp_y ($0.25 \leq x_F < 0.30$)

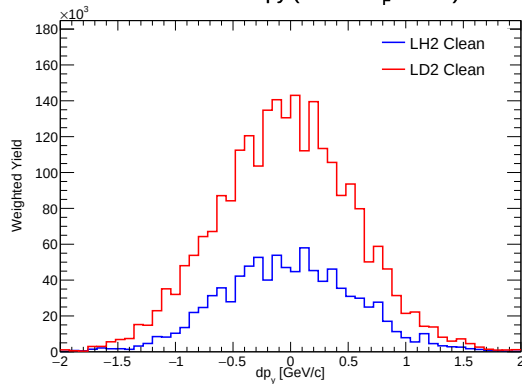


Kinematic Yields (Clean Accepted): x_F Bin 6

Clean Yield of dp_x ($0.25 \leq x_F < 0.30$)

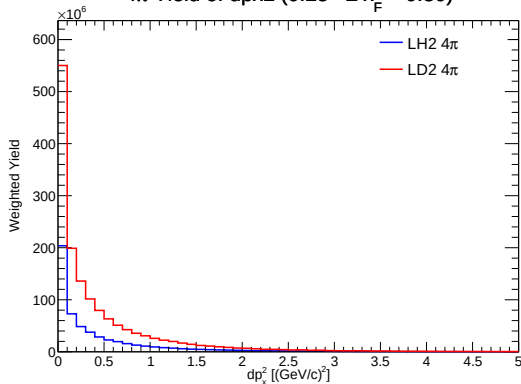


Clean Yield of dp_y ($0.25 \leq x_F < 0.30$)

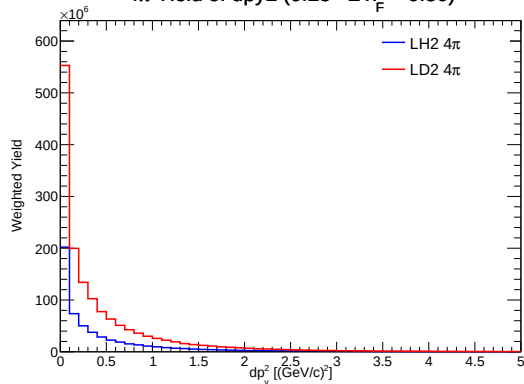


Kinematic Yields Squared (4π Thrown): x_F Bin 6

4 π Yield of dp_x^2 ($0.25 \leq x_F < 0.30$)

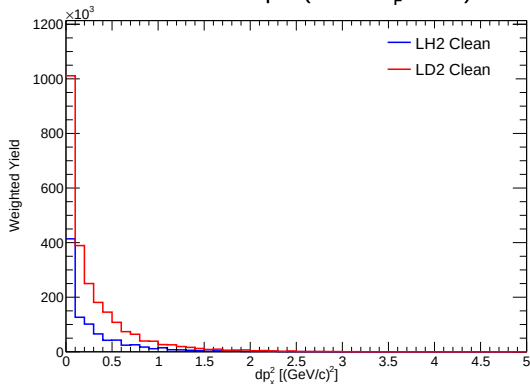


4 π Yield of dp_y^2 ($0.25 \leq x_F < 0.30$)

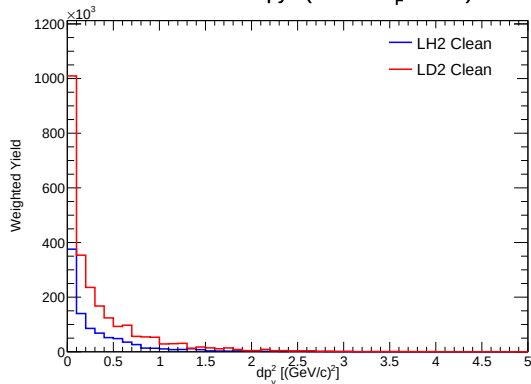


Kinematic Yields Squared (Clean Accepted): x_F Bin 6

Clean Yield of dp_x^2 ($0.25 \leq x_F < 0.30$)

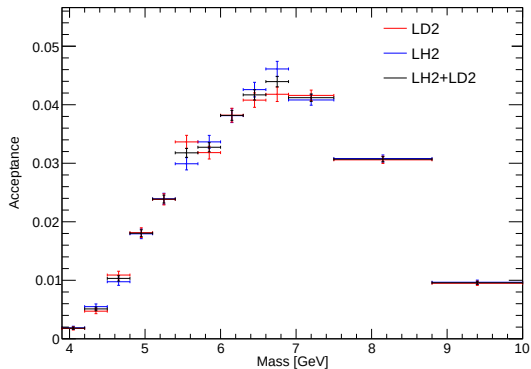


Clean Yield of dp_y^2 ($0.25 \leq x_F < 0.30$)

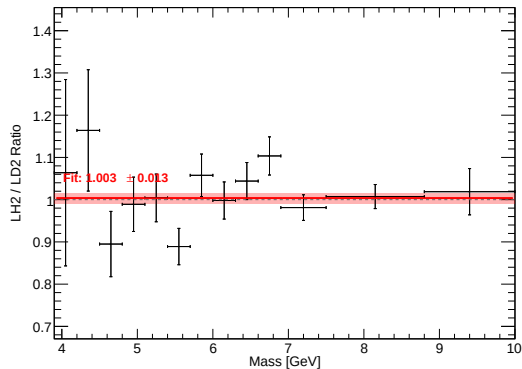


Overlay & Ratio (Mass sliced by x_F): x_F Bin 6

$0.25 \leq x_F < 0.30$

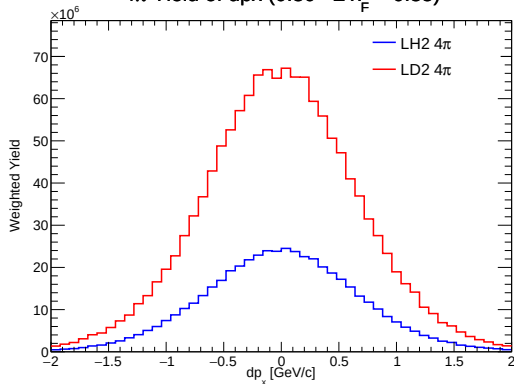


LH2/LD2 $0.25 \leq x_F < 0.30$

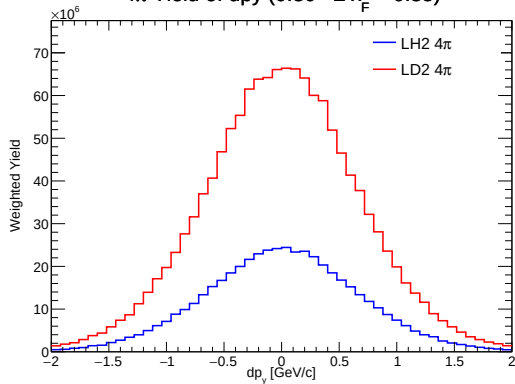


Kinematic Yields (4π Thrown): x_F Bin 7

4π Yield of dp_x ($0.30 \leq x_F < 0.35$)

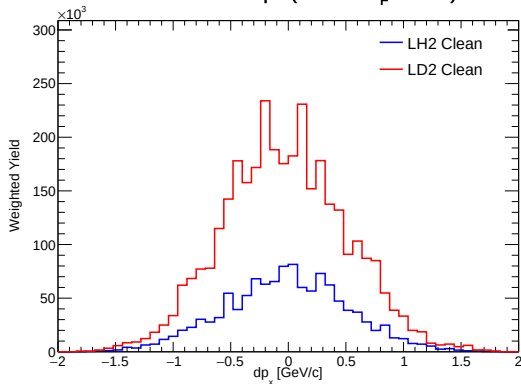


4π Yield of dp_y ($0.30 \leq x_F < 0.35$)

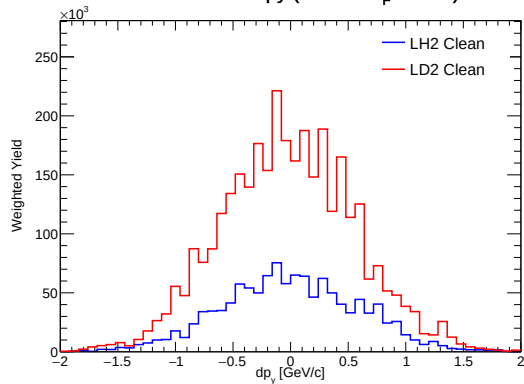


Kinematic Yields (Clean Accepted): x_F Bin 7

Clean Yield of dp_x ($0.30 \leq x_F < 0.35$)

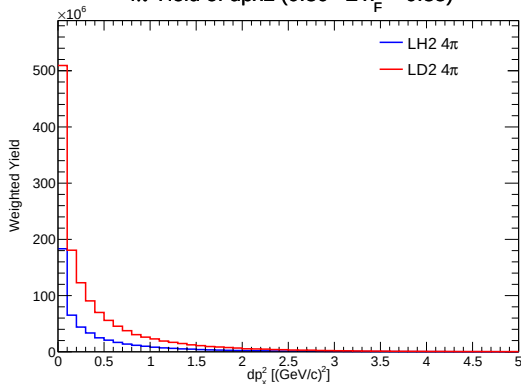


Clean Yield of dp_y ($0.30 \leq x_F < 0.35$)

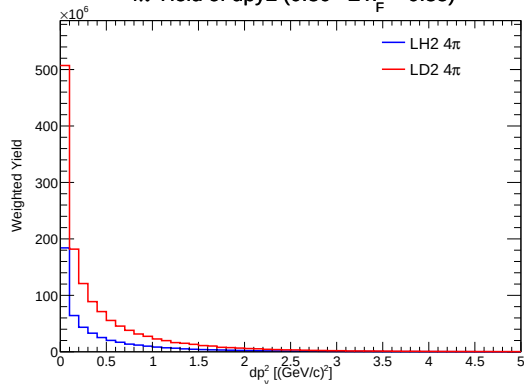


Kinematic Yields Squared (4π Thrown): x_F Bin 7

4 π Yield of dp_x^2 ($0.30 \leq x_F < 0.35$)

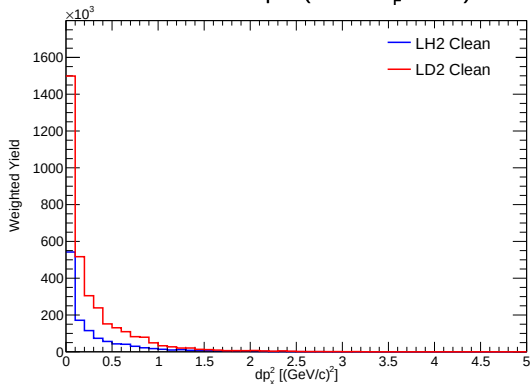


4 π Yield of dp_y^2 ($0.30 \leq x_F < 0.35$)

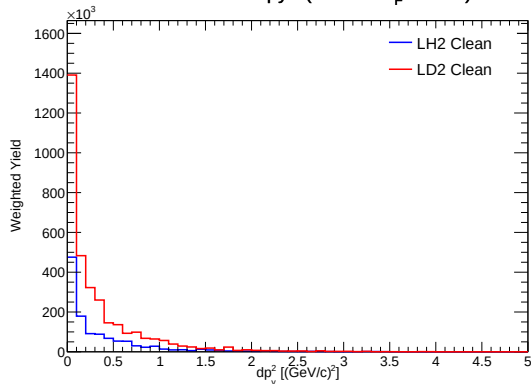


Kinematic Yields Squared (Clean Accepted): x_F Bin 7

Clean Yield of dp_x^2 ($0.30 \leq x_F < 0.35$)

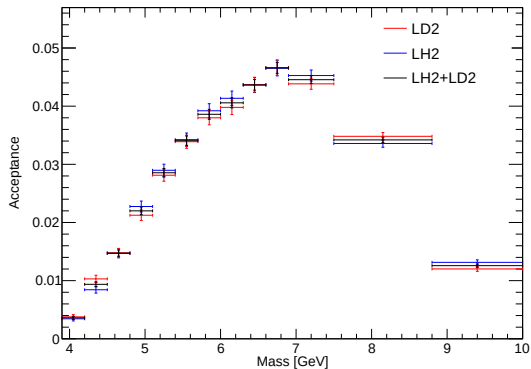


Clean Yield of dp_y^2 ($0.30 \leq x_F < 0.35$)

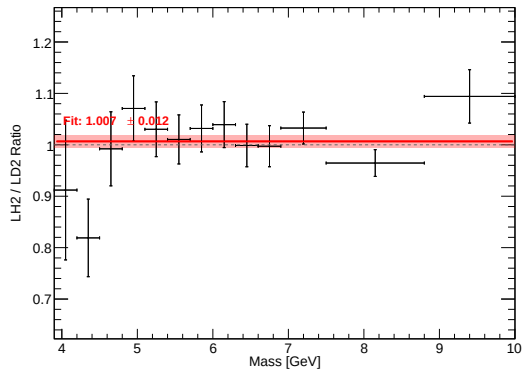


Overlay & Ratio (Mass sliced by x_F): x_F Bin 7

$0.30 \leq x_F < 0.35$

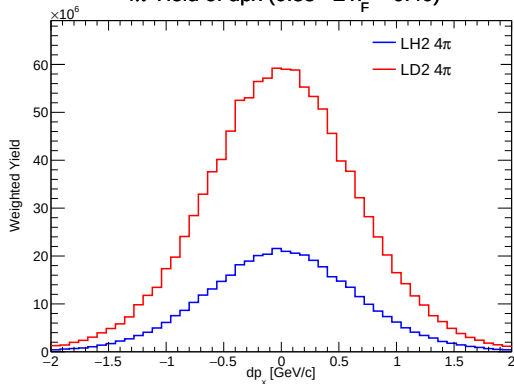


LH2/LD2 $0.30 \leq x_F < 0.35$

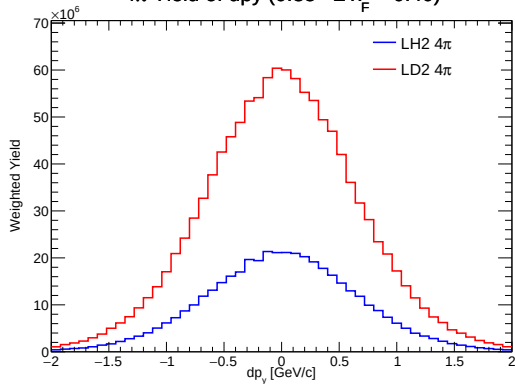


Kinematic Yields (4π Thrown): x_F Bin 8

4π Yield of dp_x ($0.35 \leq x_F < 0.40$)

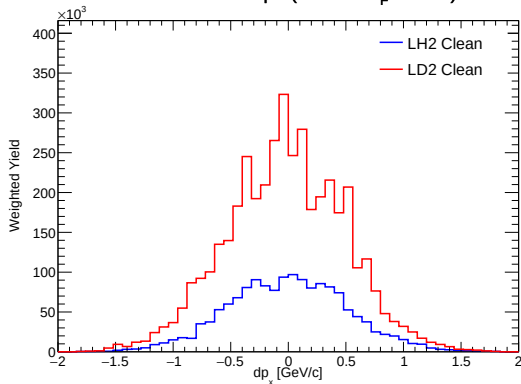


4π Yield of dp_y ($0.35 \leq x_F < 0.40$)

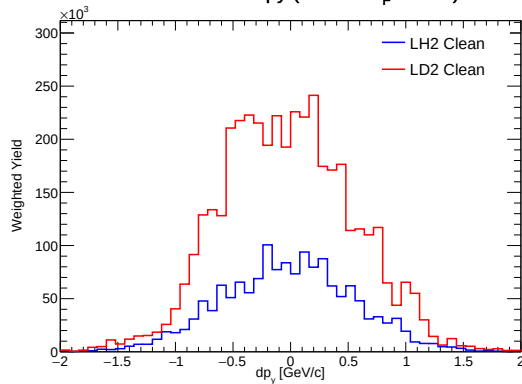


Kinematic Yields (Clean Accepted): x_F Bin 8

Clean Yield of dp_x ($0.35 \leq x_F < 0.40$)

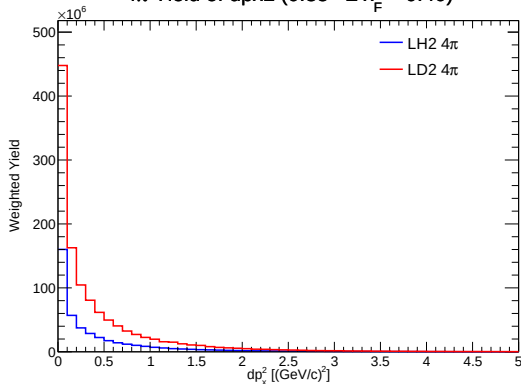


Clean Yield of dp_y ($0.35 \leq x_F < 0.40$)

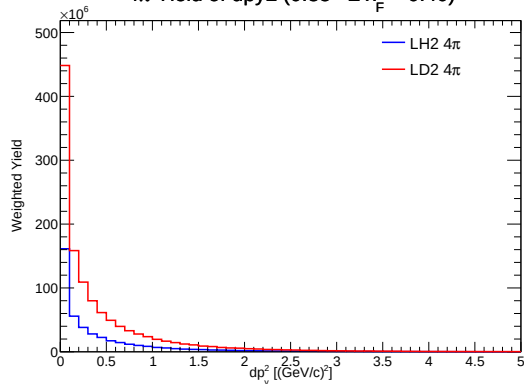


Kinematic Yields Squared (4π Thrown): x_F Bin 8

4 π Yield of dp_x^2 ($0.35 \leq x_F < 0.40$)

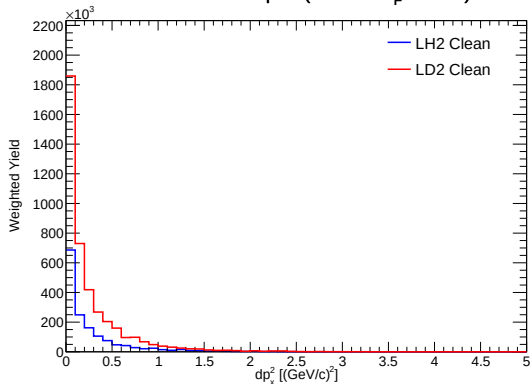


4 π Yield of dp_y^2 ($0.35 \leq x_F < 0.40$)

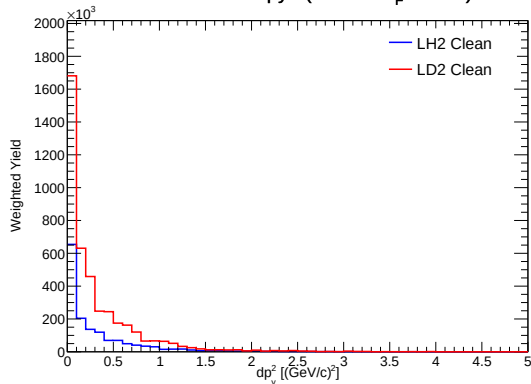


Kinematic Yields Squared (Clean Accepted): x_F Bin 8

Clean Yield of dp_x^2 ($0.35 \leq x_F < 0.40$)

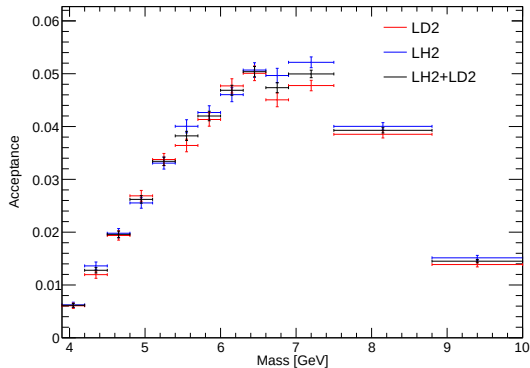


Clean Yield of dp_y^2 ($0.35 \leq x_F < 0.40$)

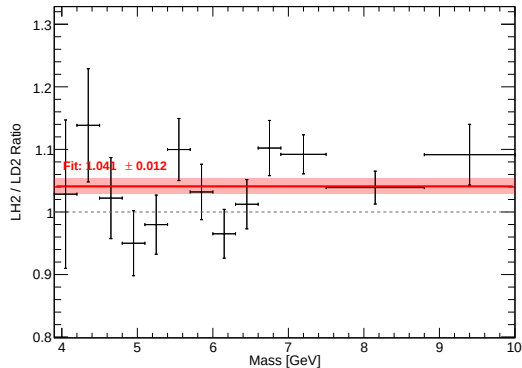


Overlay & Ratio (Mass sliced by x_F): x_F Bin 8

$0.35 \leq x_F < 0.40$

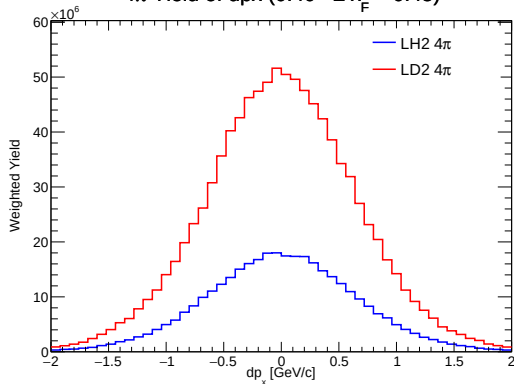


LH2/LD2 $0.35 \leq x_F < 0.40$

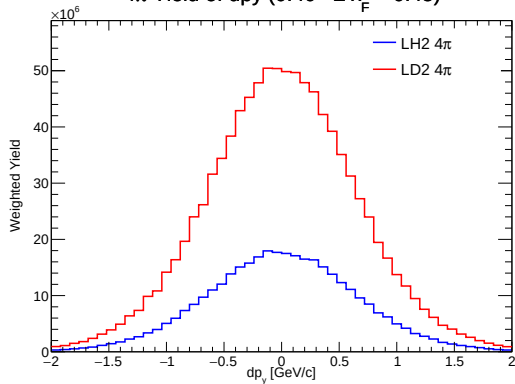


Kinematic Yields (4π Thrown): x_F Bin 9

4π Yield of dp_x ($0.40 \leq x_F < 0.45$)

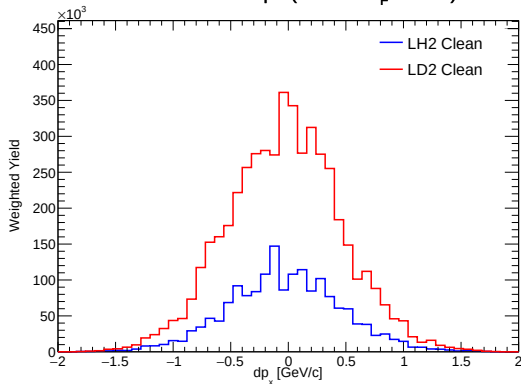


4π Yield of dp_y ($0.40 \leq x_F < 0.45$)

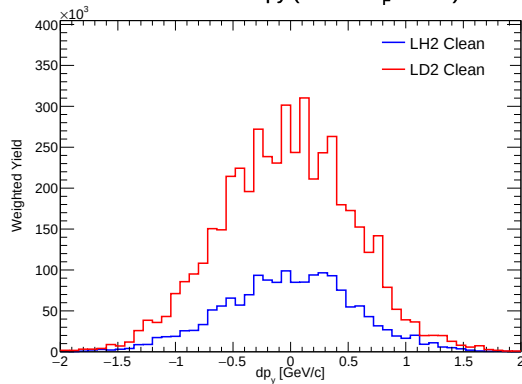


Kinematic Yields (Clean Accepted): x_F Bin 9

Clean Yield of dp_x ($0.40 \leq x_F < 0.45$)

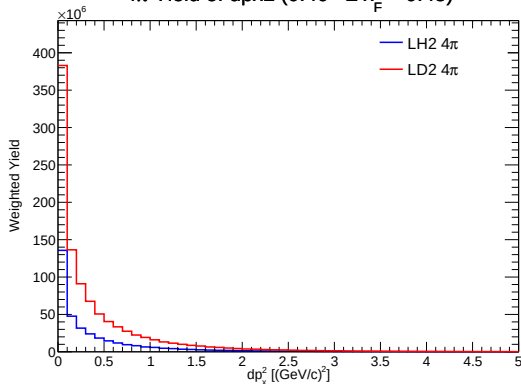


Clean Yield of dp_y ($0.40 \leq x_F < 0.45$)

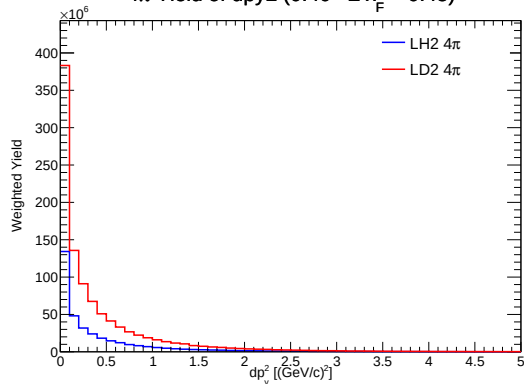


Kinematic Yields Squared (4π Thrown): x_F Bin 9

4 π Yield of dp_x^2 ($0.40 \leq x_F < 0.45$)

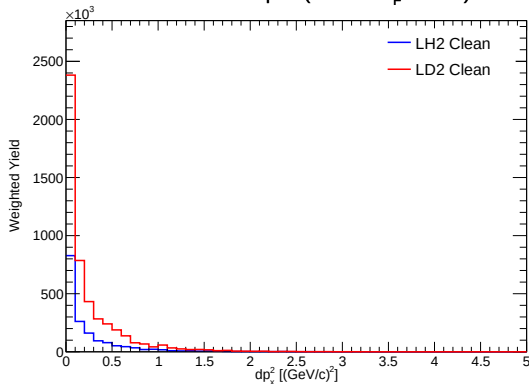


4 π Yield of dp_y^2 ($0.40 \leq x_F < 0.45$)

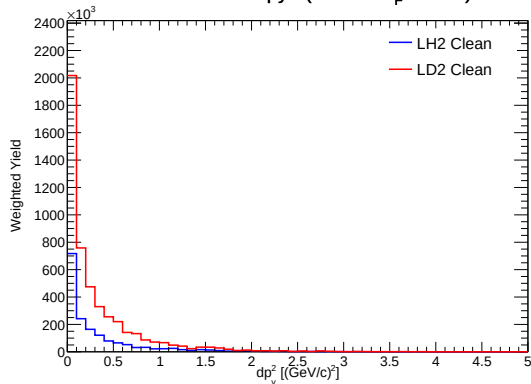


Kinematic Yields Squared (Clean Accepted): x_F Bin 9

Clean Yield of dp_x^2 ($0.40 \leq x_F < 0.45$)

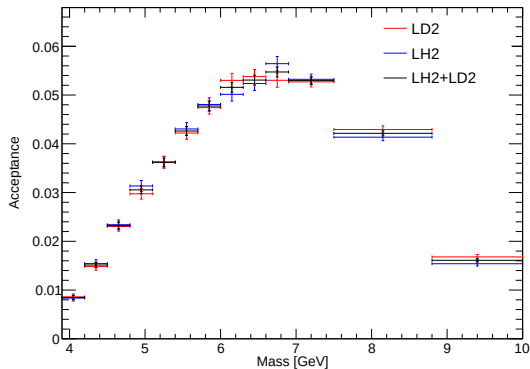


Clean Yield of dp_y^2 ($0.40 \leq x_F < 0.45$)

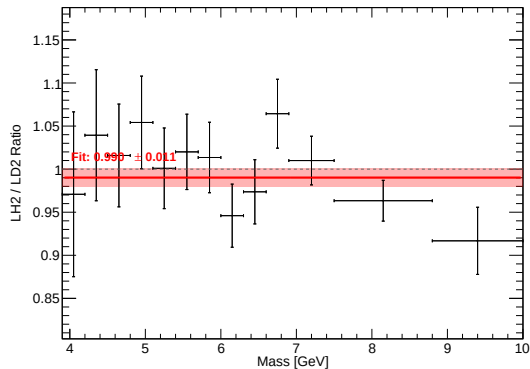


Overlay & Ratio (Mass sliced by x_F): x_F Bin 9

$0.40 \leq x_F < 0.45$

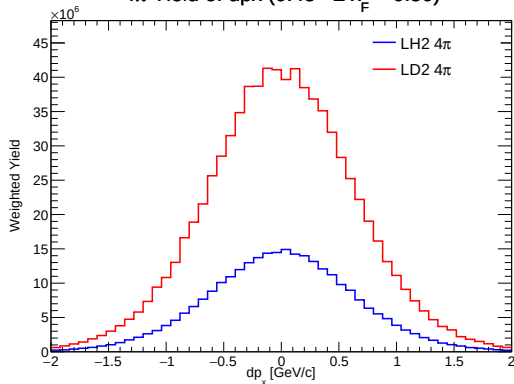


LH2/LD2 $0.40 \leq x_F < 0.45$

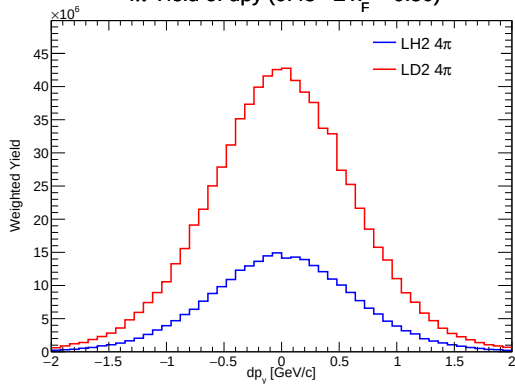


Kinematic Yields (4π Thrown): x_F Bin 10

4 π Yield of dp_x ($0.45 \leq x_F < 0.50$)

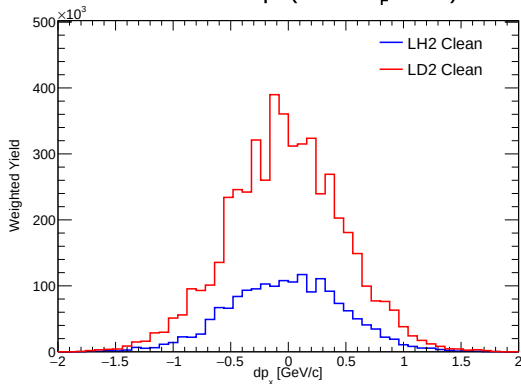


4 π Yield of dp_y ($0.45 \leq x_F < 0.50$)

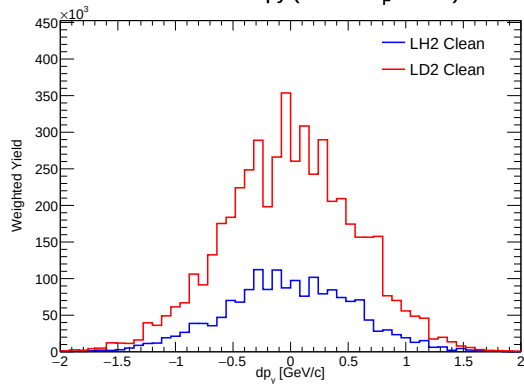


Kinematic Yields (Clean Accepted): x_F Bin 10

Clean Yield of dp_x ($0.45 \leq x_F < 0.50$)

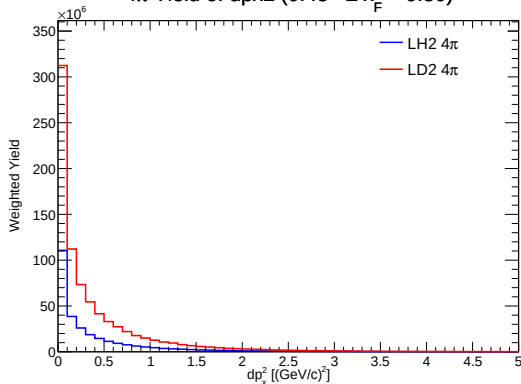


Clean Yield of dp_y ($0.45 \leq x_F < 0.50$)

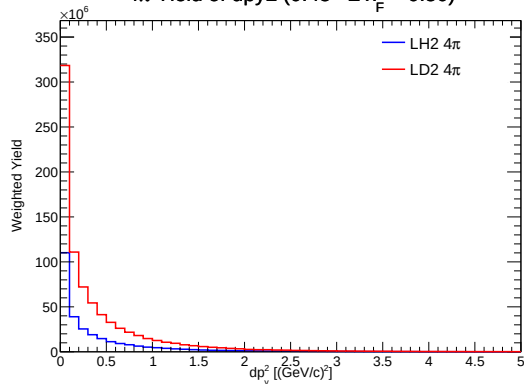


Kinematic Yields Squared (4π Thrown): x_F Bin 10

4 π Yield of dp_x^2 ($0.45 \leq x_F < 0.50$)

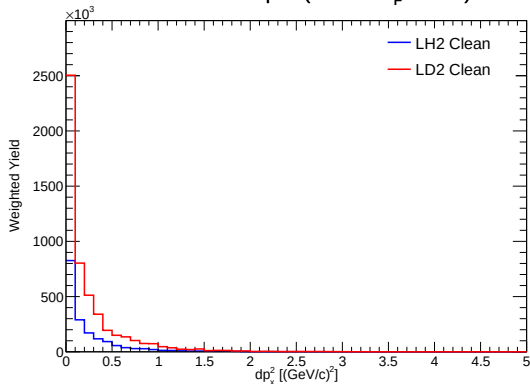


4 π Yield of dp_y^2 ($0.45 \leq x_F < 0.50$)

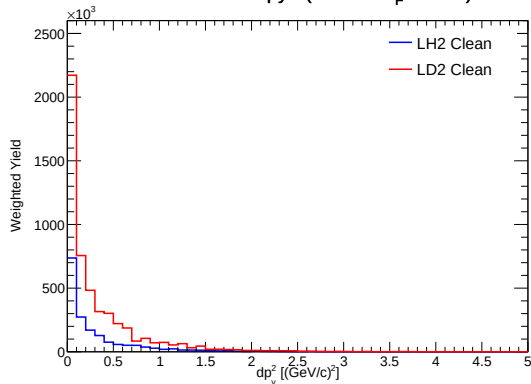


Kinematic Yields Squared (Clean Accepted): x_F Bin 10

Clean Yield of dp_x^2 ($0.45 \leq x_F < 0.50$)

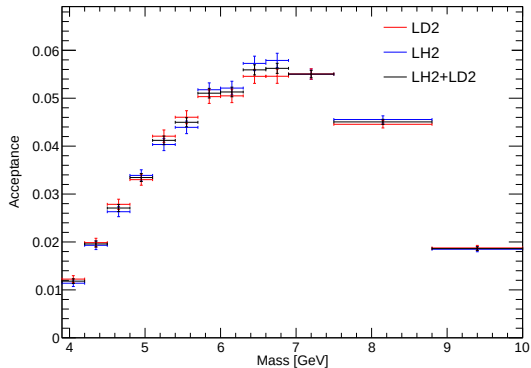


Clean Yield of dp_y^2 ($0.45 \leq x_F < 0.50$)

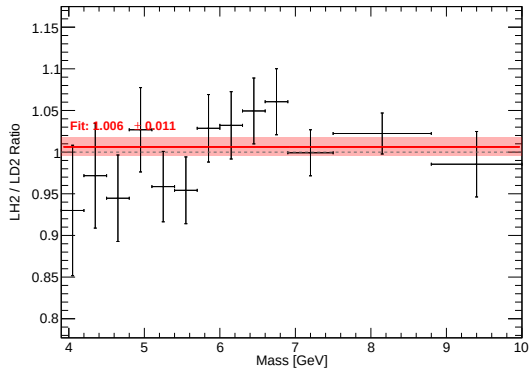


Overlay & Ratio (Mass sliced by x_F): x_F Bin 10

$0.45 \leq x_F < 0.50$

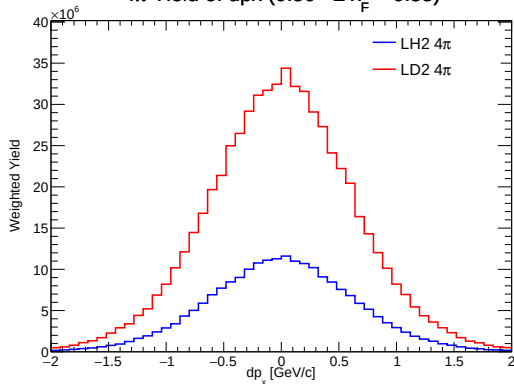


LH2/LD2 $0.45 \leq x_F < 0.50$

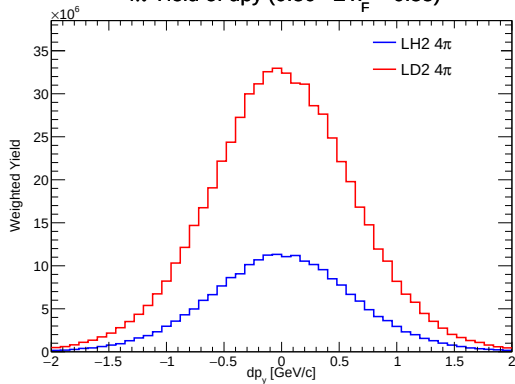


Kinematic Yields (4π Thrown): x_F Bin 11

4π Yield of dp_x ($0.50 \leq x_F < 0.55$)

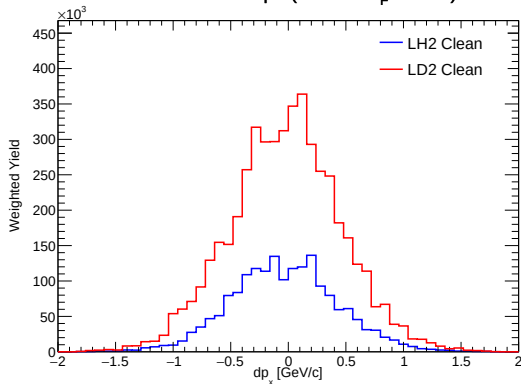


4π Yield of dp_y ($0.50 \leq x_F < 0.55$)

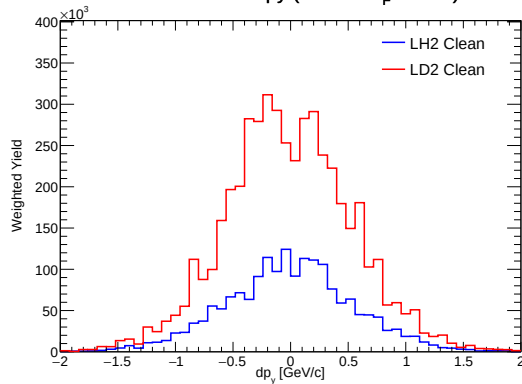


Kinematic Yields (Clean Accepted): x_F Bin 11

Clean Yield of dp_x ($0.50 \leq x_F < 0.55$)

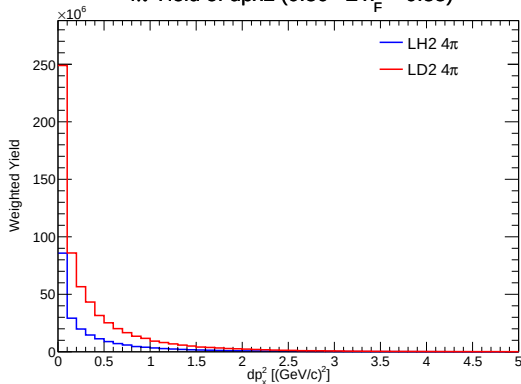


Clean Yield of dp_y ($0.50 \leq x_F < 0.55$)

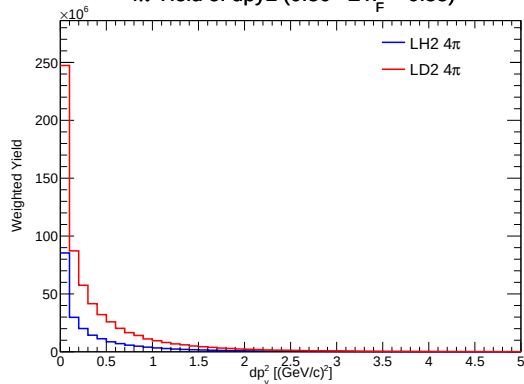


Kinematic Yields Squared (4π Thrown): x_F Bin 11

4 π Yield of dp_x^2 ($0.50 \leq x_F < 0.55$)

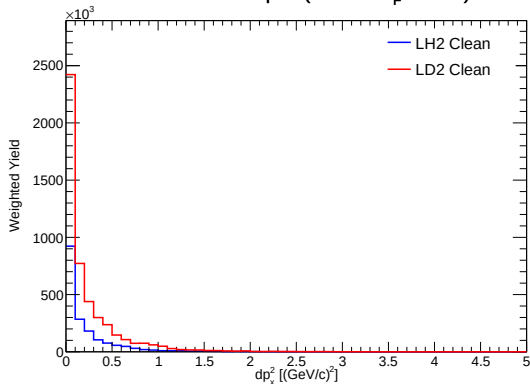


4 π Yield of dp_y^2 ($0.50 \leq x_F < 0.55$)

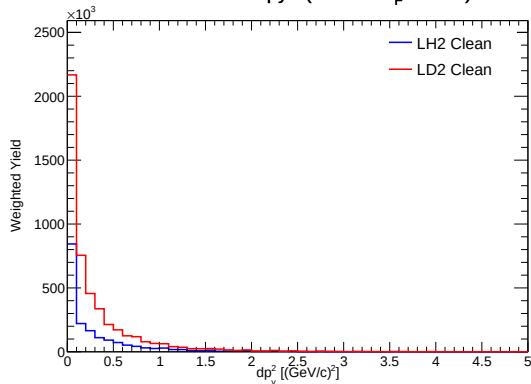


Kinematic Yields Squared (Clean Accepted): x_F Bin 11

Clean Yield of dp_x^2 ($0.50 \leq x_F < 0.55$)

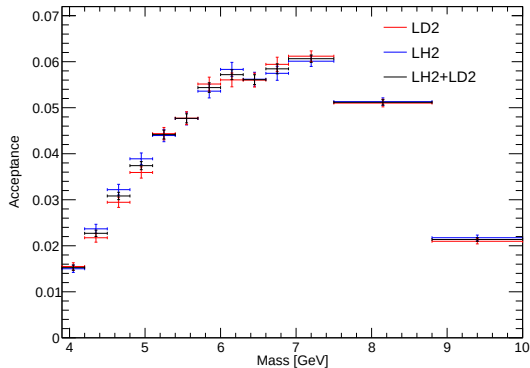


Clean Yield of dp_y^2 ($0.50 \leq x_F < 0.55$)

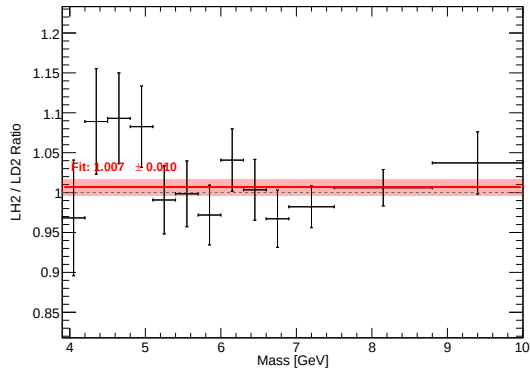


Overlay & Ratio (Mass sliced by x_F): x_F Bin 11

$0.50 \leq x_F < 0.55$

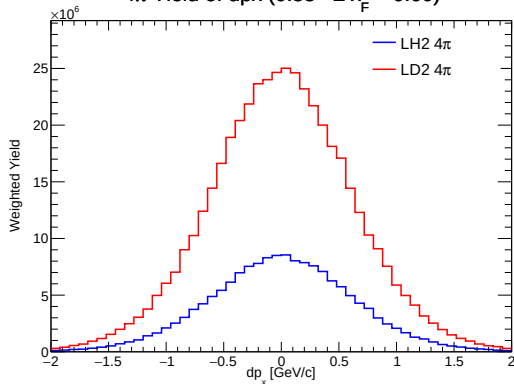


LH2/LD2 $0.50 \leq x_F < 0.55$

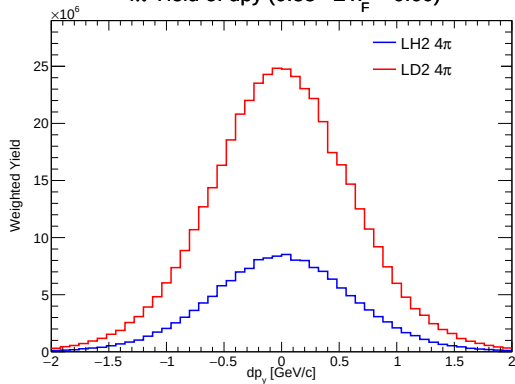


Kinematic Yields (4π Thrown): x_F Bin 12

4π Yield of dp_x ($0.55 \leq x_F < 0.60$)

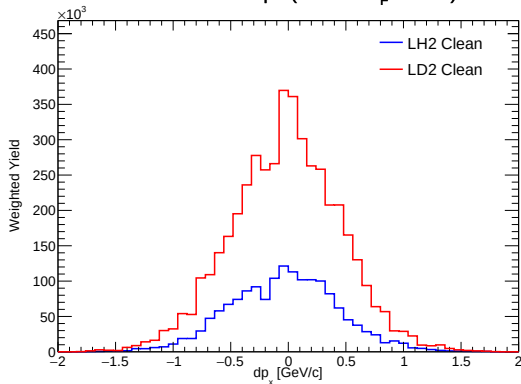


4π Yield of dp_y ($0.55 \leq x_F < 0.60$)

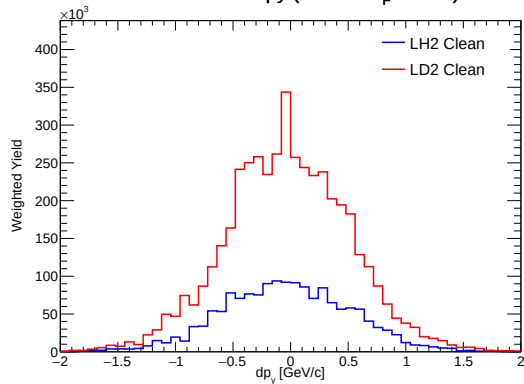


Kinematic Yields (Clean Accepted): x_F Bin 12

Clean Yield of dp_x ($0.55 \leq x_F < 0.60$)

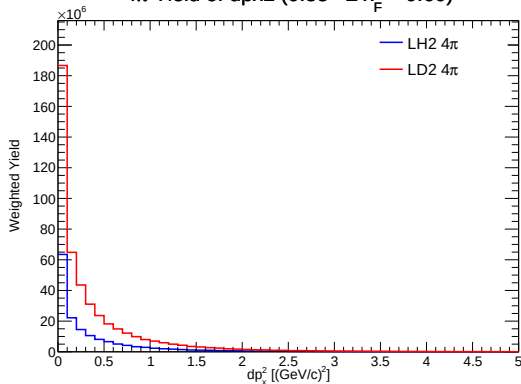


Clean Yield of dp_y ($0.55 \leq x_F < 0.60$)

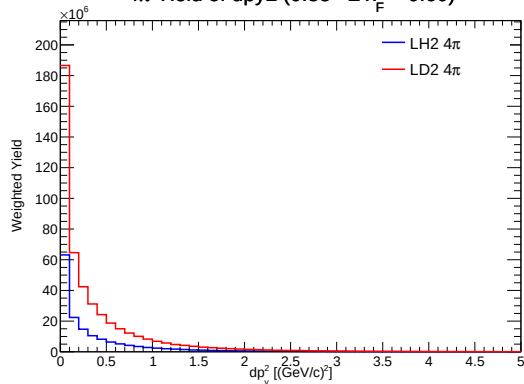


Kinematic Yields Squared (4π Thrown): x_F Bin 12

4 π Yield of dp_x^2 ($0.55 \leq x_F < 0.60$)

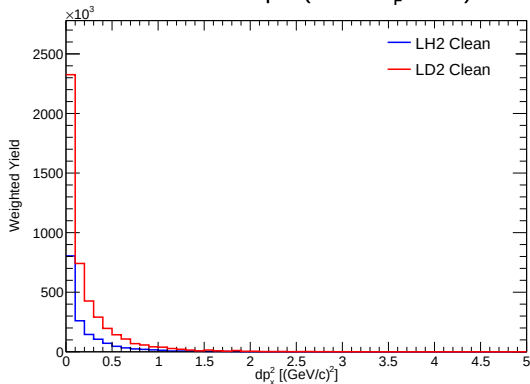


4 π Yield of dp_y^2 ($0.55 \leq x_F < 0.60$)

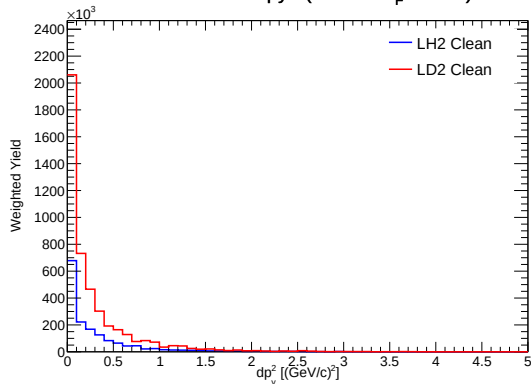


Kinematic Yields Squared (Clean Accepted): x_F Bin 12

Clean Yield of dp_x^2 ($0.55 \leq x_F < 0.60$)

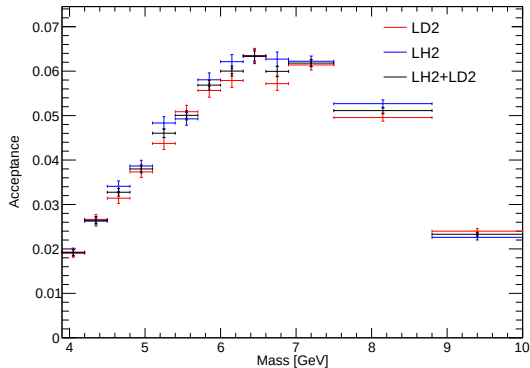


Clean Yield of dp_y^2 ($0.55 \leq x_F < 0.60$)

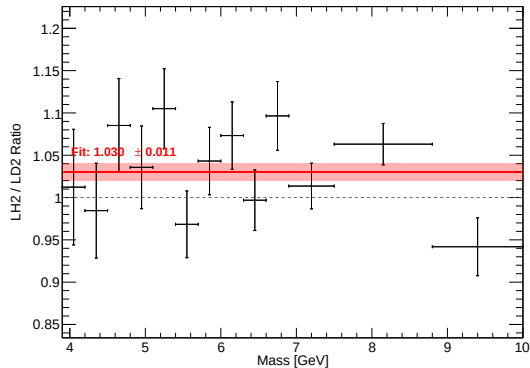


Overlay & Ratio (Mass sliced by x_F): x_F Bin 12

$0.55 \leq x_F < 0.60$

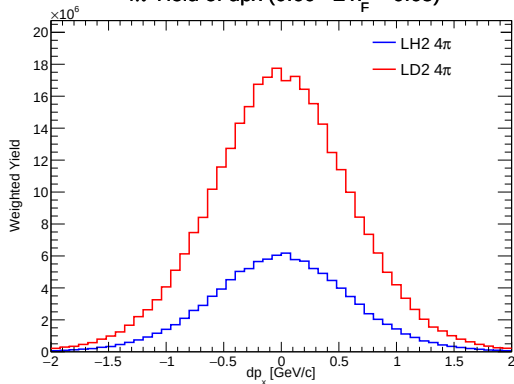


LH2/LD2 $0.55 \leq x_F < 0.60$

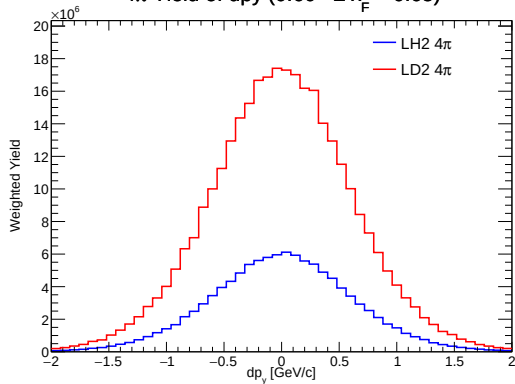


Kinematic Yields (4π Thrown): x_F Bin 13

4 π Yield of dp_x ($0.60 \leq x_F < 0.65$)

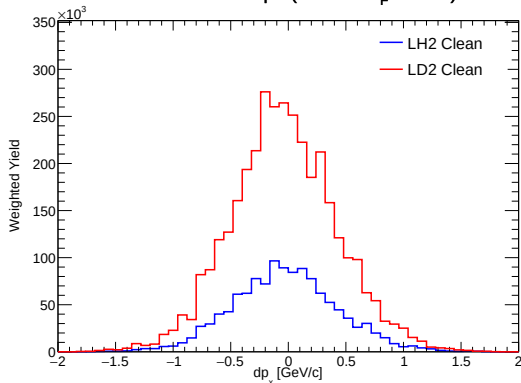


4 π Yield of dp_y ($0.60 \leq x_F < 0.65$)

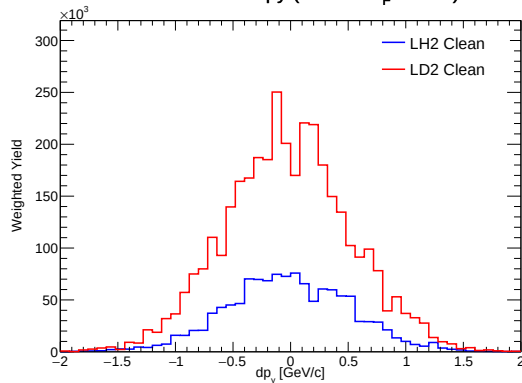


Kinematic Yields (Clean Accepted): x_F Bin 13

Clean Yield of dp_x ($0.60 \leq x_F < 0.65$)

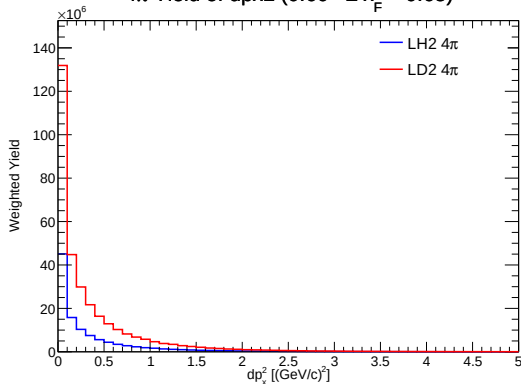


Clean Yield of dp_y ($0.60 \leq x_F < 0.65$)

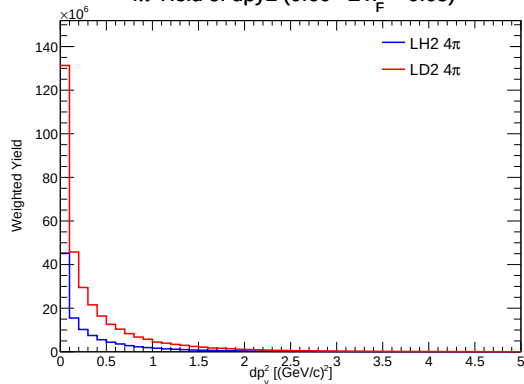


Kinematic Yields Squared (4π Thrown): x_F Bin 13

4 π Yield of dp_x^2 ($0.60 \leq x_F < 0.65$)

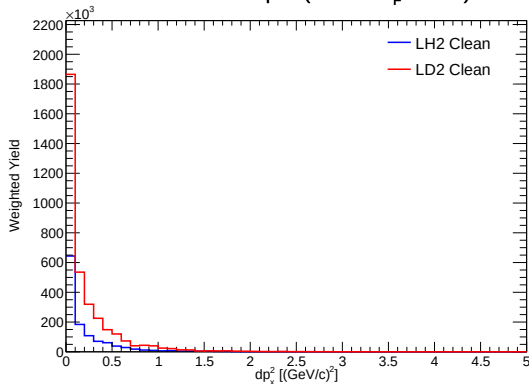


4 π Yield of dp_y^2 ($0.60 \leq x_F < 0.65$)

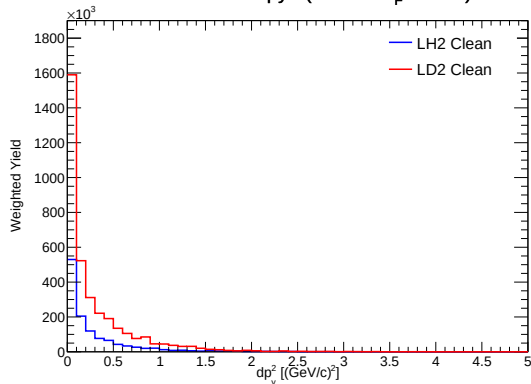


Kinematic Yields Squared (Clean Accepted): x_F Bin 13

Clean Yield of dp_x^2 ($0.60 \leq x_F < 0.65$)

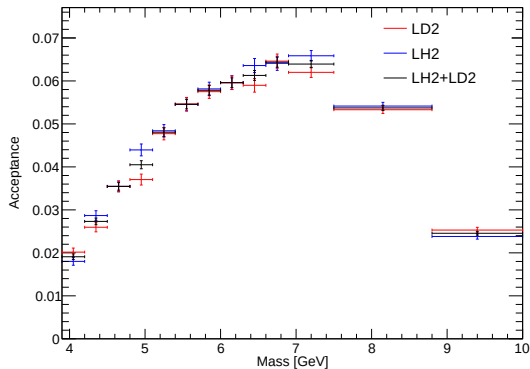


Clean Yield of dp_y^2 ($0.60 \leq x_F < 0.65$)

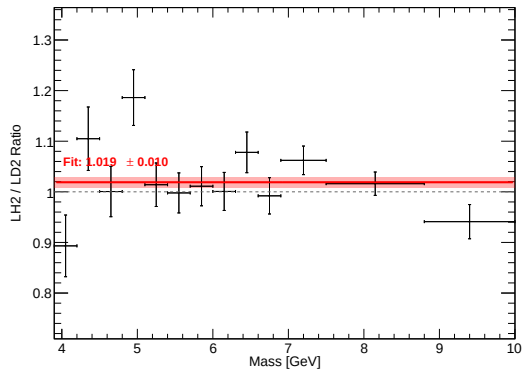


Overlay & Ratio (Mass sliced by x_F): x_F Bin 13

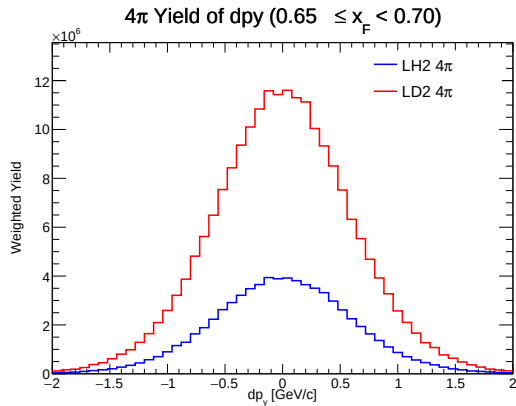
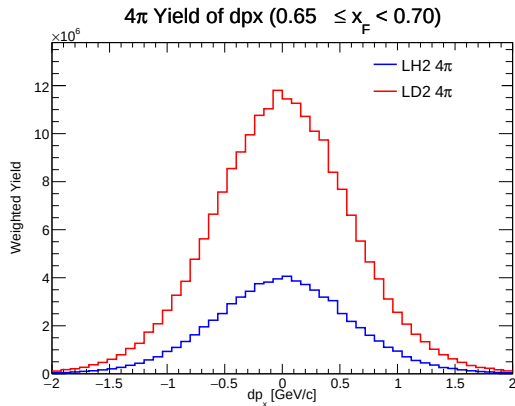
$0.60 \leq x_F < 0.65$



LH2/LD2 $0.60 \leq x_F < 0.65$

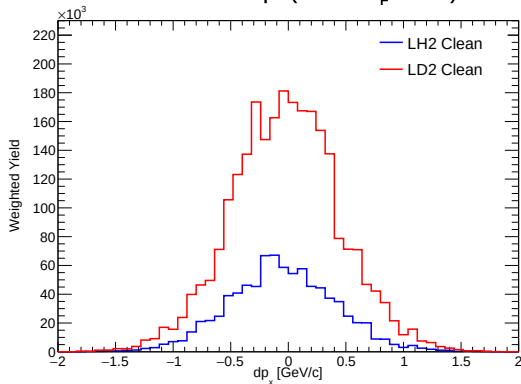


Kinematic Yields (4π Thrown): x_F Bin 14

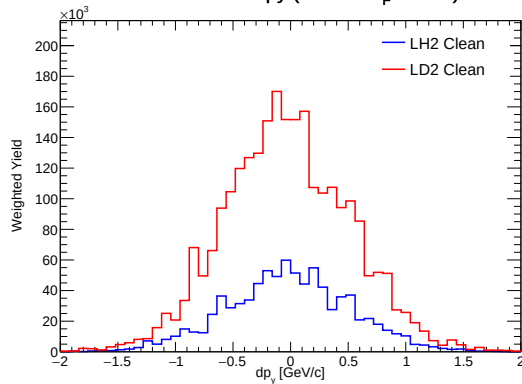


Kinematic Yields (Clean Accepted): x_F Bin 14

Clean Yield of dp_x ($0.65 \leq x_F < 0.70$)

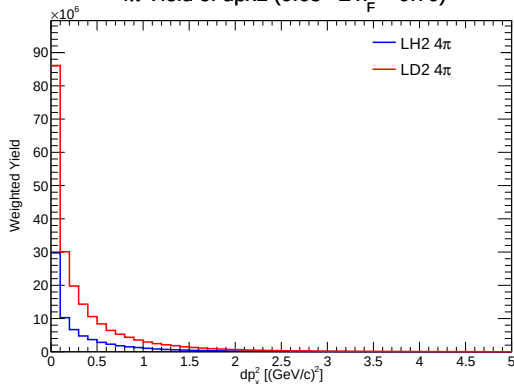


Clean Yield of dp_y ($0.65 \leq x_F < 0.70$)

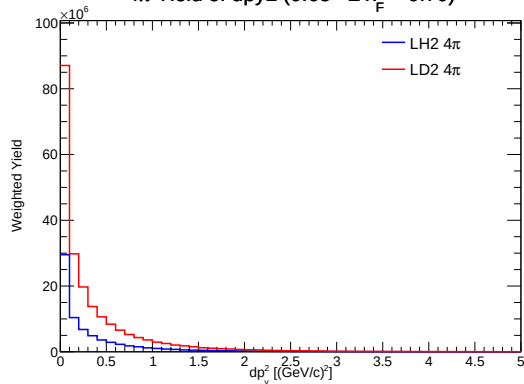


Kinematic Yields Squared (4π Thrown): x_F Bin 14

4 π Yield of dp_x^2 ($0.65 \leq x_F < 0.70$)

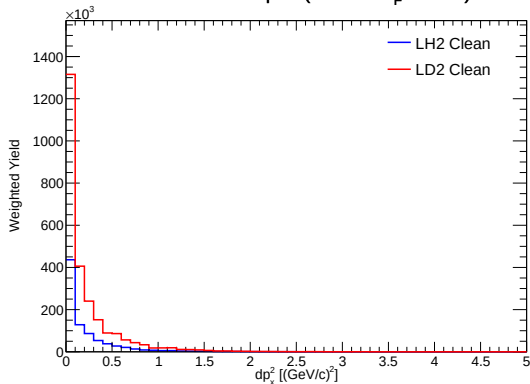


4 π Yield of dp_y^2 ($0.65 \leq x_F < 0.70$)

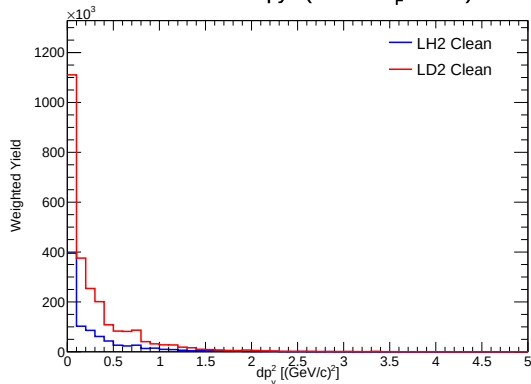


Kinematic Yields Squared (Clean Accepted): x_F Bin 14

Clean Yield of dp_x^2 ($0.65 \leq x_F < 0.70$)

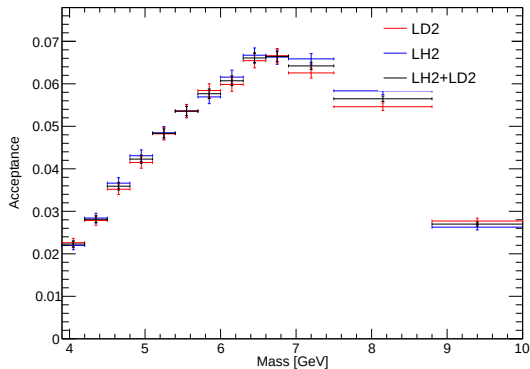


Clean Yield of dp_y^2 ($0.65 \leq x_F < 0.70$)

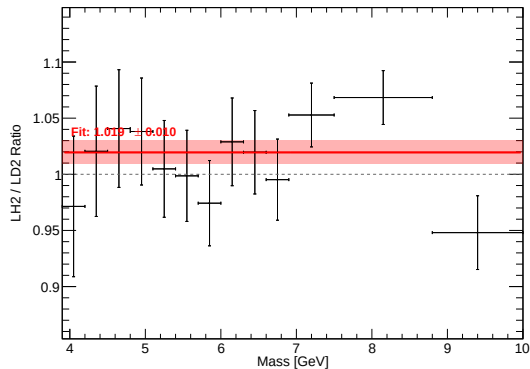


Overlay & Ratio (Mass sliced by x_F): x_F Bin 14

$0.65 \leq x_F < 0.70$

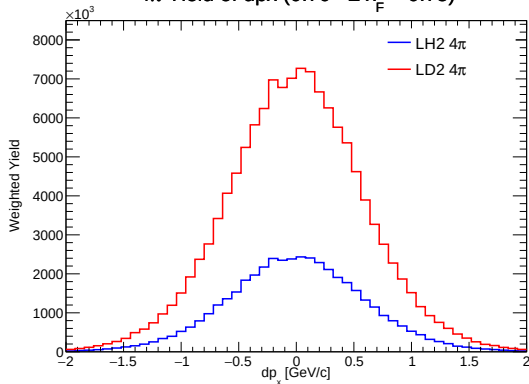


LH2/LD2 $0.65 \leq x_F < 0.70$

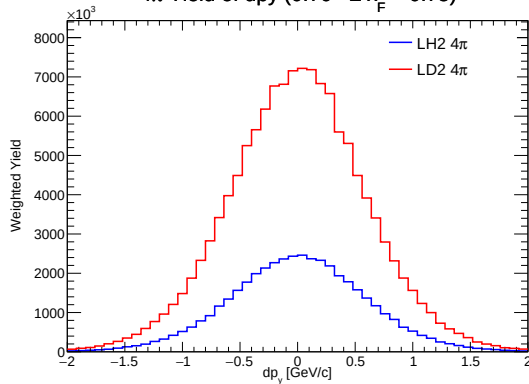


Kinematic Yields (4π Thrown): x_F Bin 15

4 π Yield of dp_x ($0.70 \leq x_F < 0.75$)

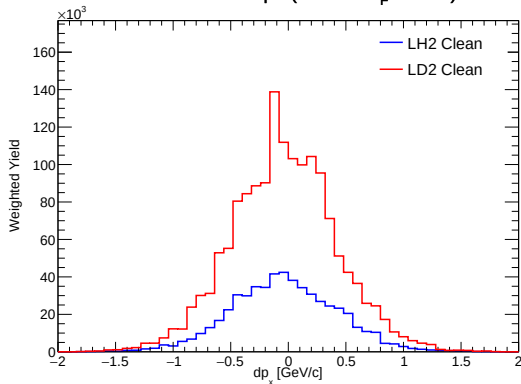


4 π Yield of dp_y ($0.70 \leq x_F < 0.75$)

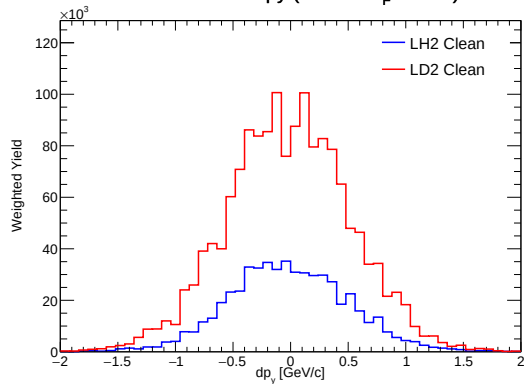


Kinematic Yields (Clean Accepted): x_F Bin 15

Clean Yield of dp_x ($0.70 \leq x_F < 0.75$)

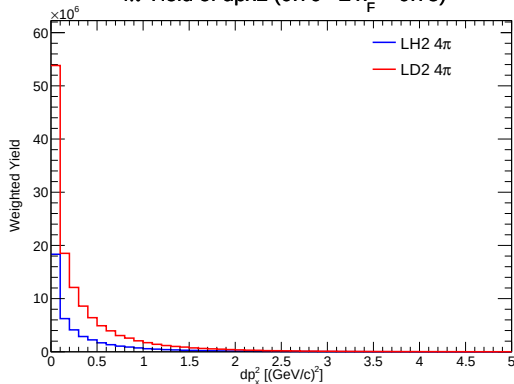


Clean Yield of dp_y ($0.70 \leq x_F < 0.75$)

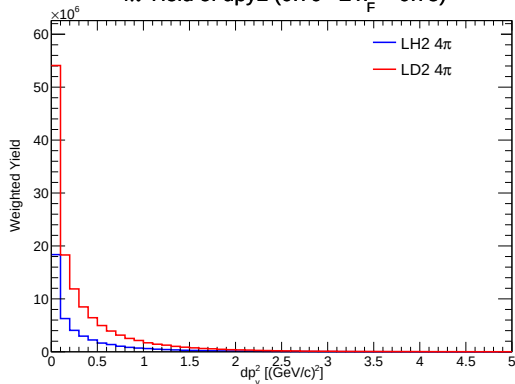


Kinematic Yields Squared (4π Thrown): x_F Bin 15

4 π Yield of dp_x^2 ($0.70 \leq x_F < 0.75$)

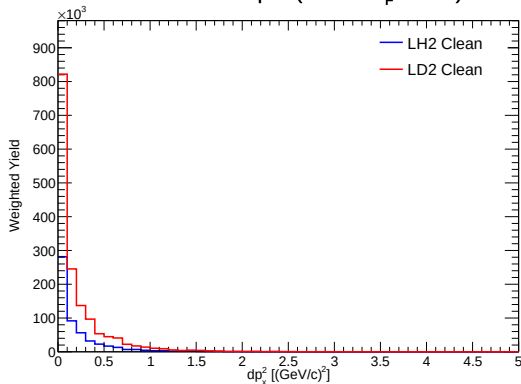


4 π Yield of dp_y^2 ($0.70 \leq x_F < 0.75$)

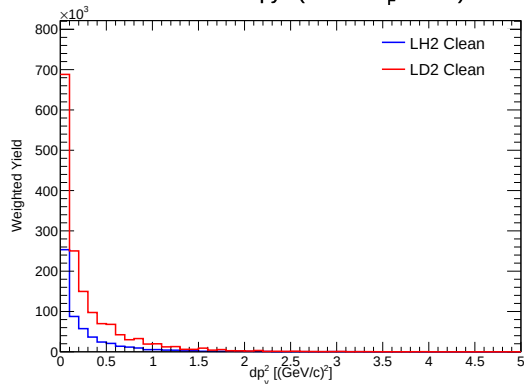


Kinematic Yields Squared (Clean Accepted): x_F Bin 15

Clean Yield of dp_x^2 ($0.70 \leq x_F < 0.75$)

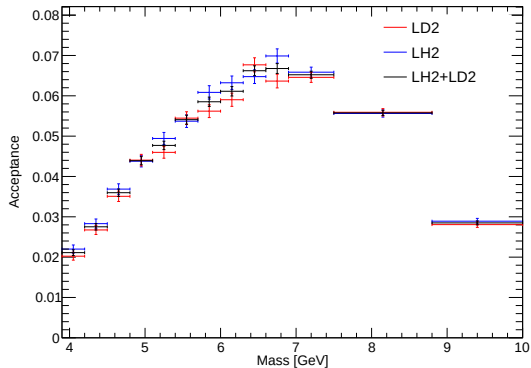


Clean Yield of dp_y^2 ($0.70 \leq x_F < 0.75$)

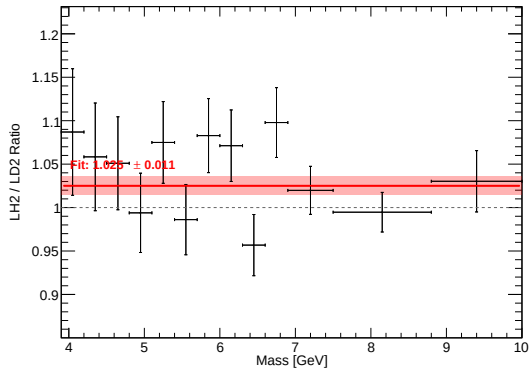


Overlay & Ratio (Mass sliced by x_F): x_F Bin 15

$0.70 \leq x_F < 0.75$

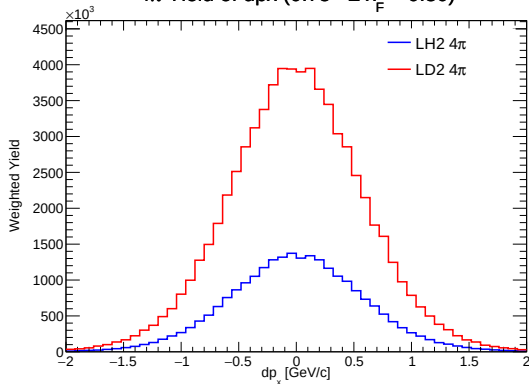


LH2/LD2 $0.70 \leq x_F < 0.75$

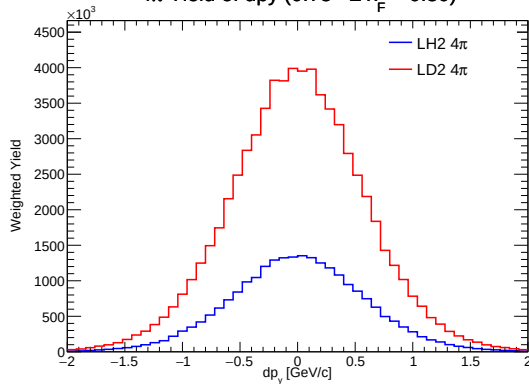


Kinematic Yields (4π Thrown): x_F Bin 16

4π Yield of dp_x ($0.75 \leq x_F < 0.80$)

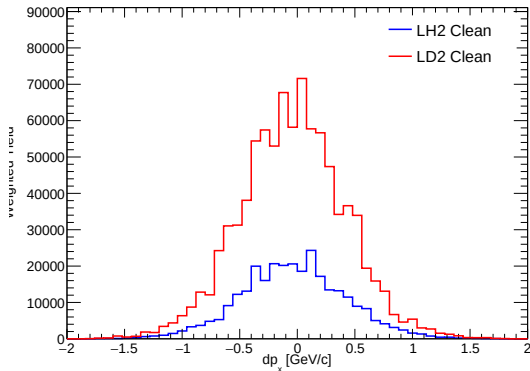


4π Yield of dp_y ($0.75 \leq x_F < 0.80$)

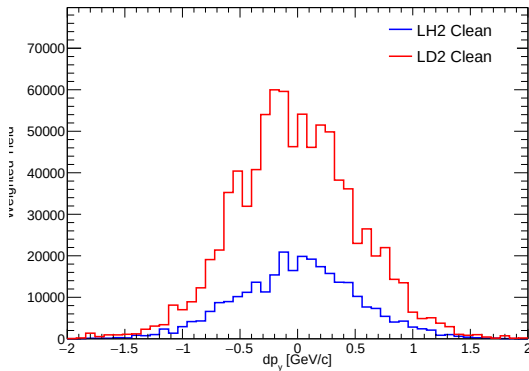


Kinematic Yields (Clean Accepted): x_F Bin 16

Clean Yield of dp_x ($0.75 \leq x_F < 0.80$)

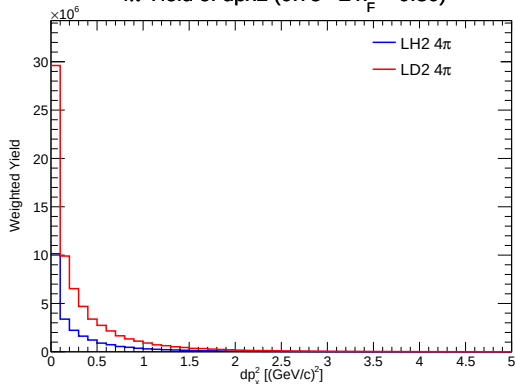


Clean Yield of dp_y ($0.75 \leq x_F < 0.80$)

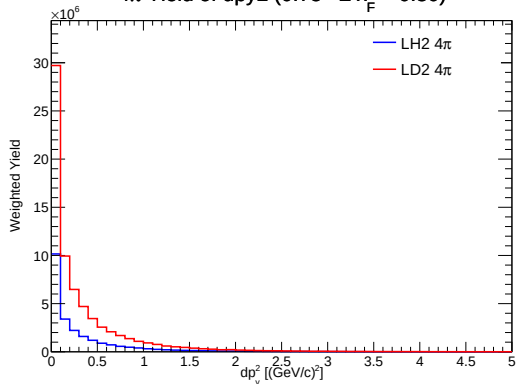


Kinematic Yields Squared (4π Thrown): x_F Bin 16

4 π Yield of dp_x^2 ($0.75 \leq x_F < 0.80$)

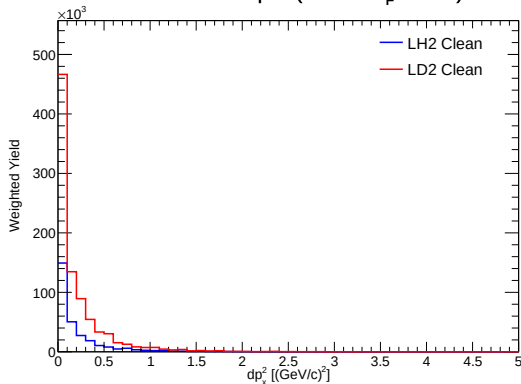


4 π Yield of dp_y^2 ($0.75 \leq x_F < 0.80$)

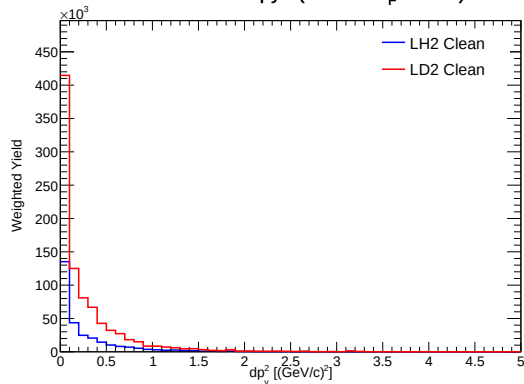


Kinematic Yields Squared (Clean Accepted): x_F Bin 16

Clean Yield of dp_x^2 ($0.75 \leq x_F < 0.80$)

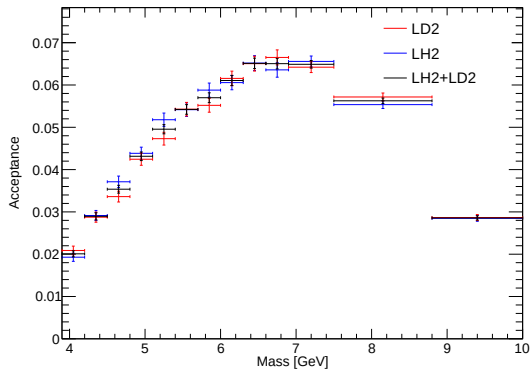


Clean Yield of dp_y^2 ($0.75 \leq x_F < 0.80$)

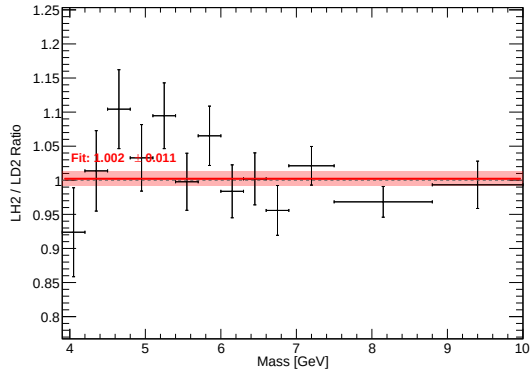


Overlay & Ratio (Mass sliced by x_F): x_F Bin 16

$0.75 \leq x_F < 0.80$

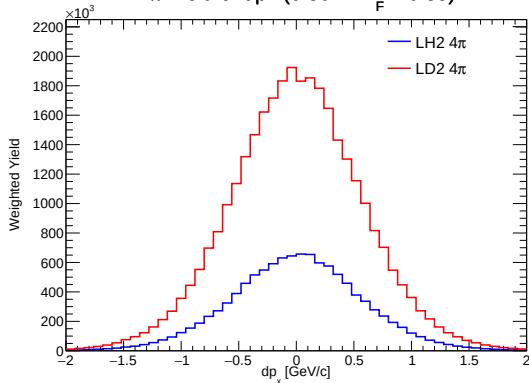


LH2/LD2 $0.75 \leq x_F < 0.80$

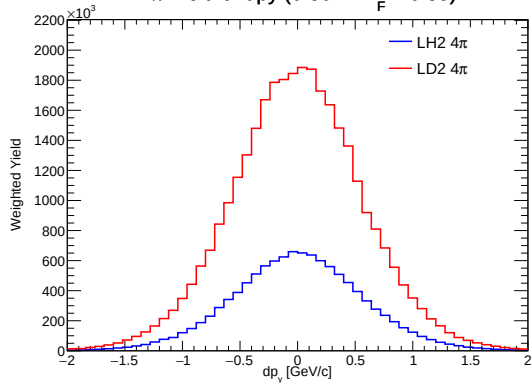


Kinematic Yields (4π Thrown): x_F Bin 17

4π Yield of dp_x ($0.80 \leq x_F < 0.85$)

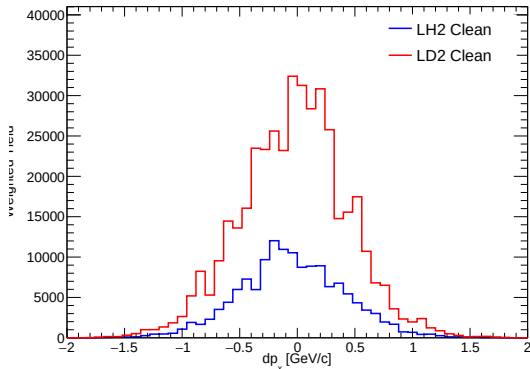


4π Yield of dp_y ($0.80 \leq x_F < 0.85$)

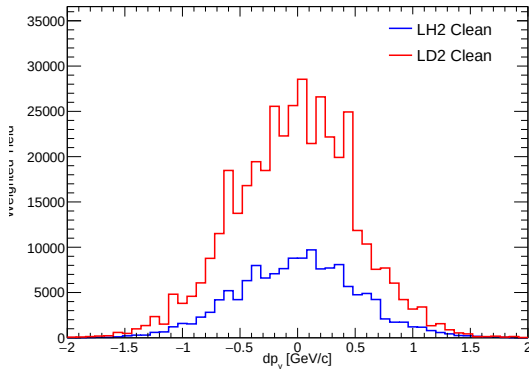


Kinematic Yields (Clean Accepted): x_F Bin 17

Clean Yield of dp_x ($0.80 \leq x_F < 0.85$)

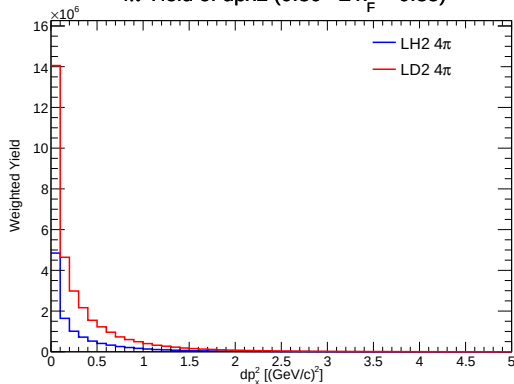


Clean Yield of dp_y ($0.80 \leq x_F < 0.85$)

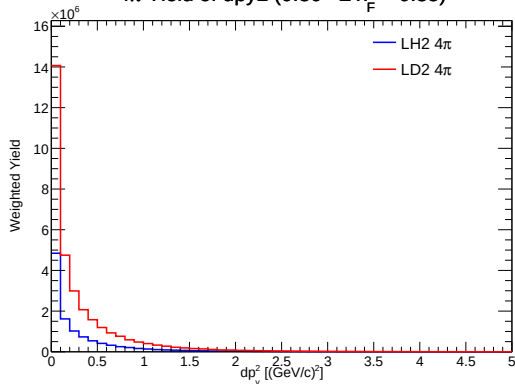


Kinematic Yields Squared (4π Thrown): x_F Bin 17

4 π Yield of dp_x^2 ($0.80 \leq x_F < 0.85$)

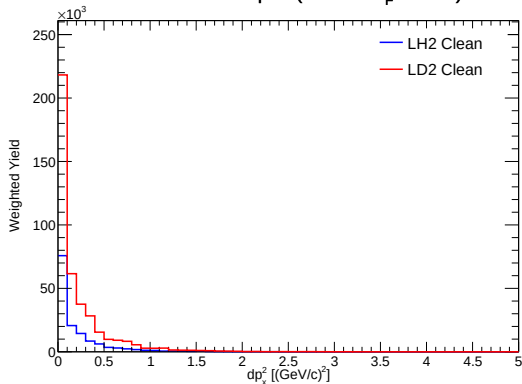


4 π Yield of dp_y^2 ($0.80 \leq x_F < 0.85$)

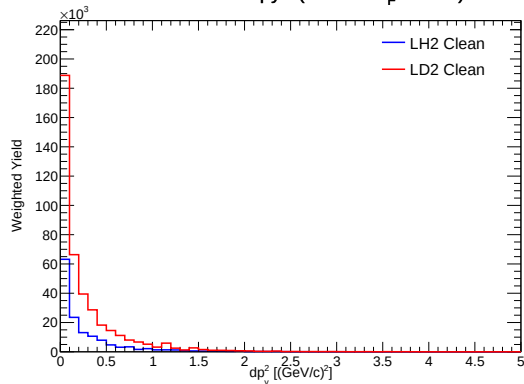


Kinematic Yields Squared (Clean Accepted): x_F Bin 17

Clean Yield of dpx2 ($0.80 \leq x_F < 0.85$)

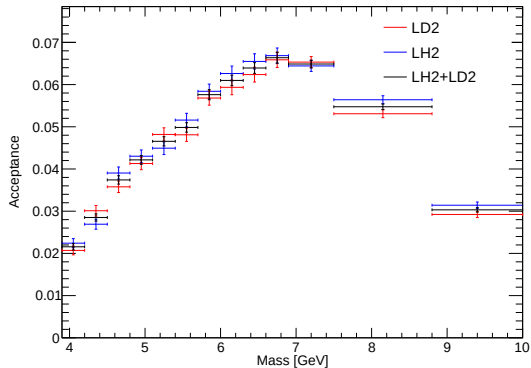


Clean Yield of dpy2 ($0.80 \leq x_F < 0.85$)

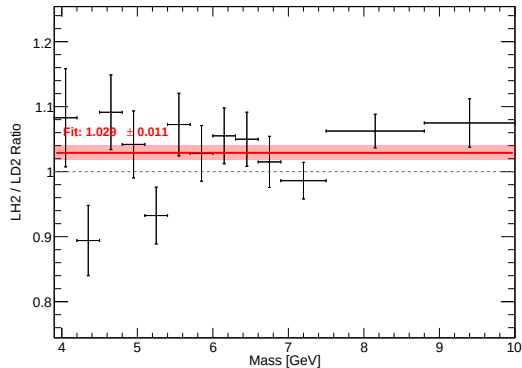


Overlay & Ratio (Mass sliced by x_F): x_F Bin 17

$0.80 \leq x_F < 0.85$

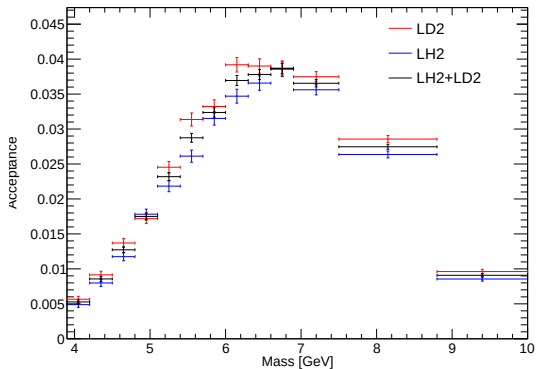


LH2/LD2 $0.80 \leq x_F < 0.85$

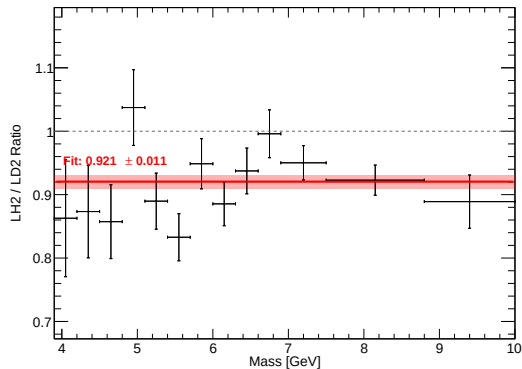


Overlay & Ratio (Mass sliced by p_T): p_T Bin 0

$0.00 \leq p_T < 0.32$

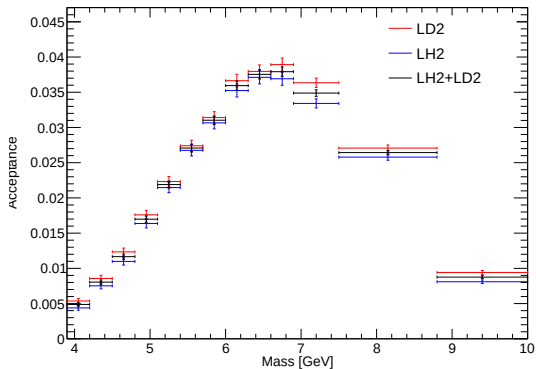


LH2/LD2 $0.00 \leq p_T < 0.32$

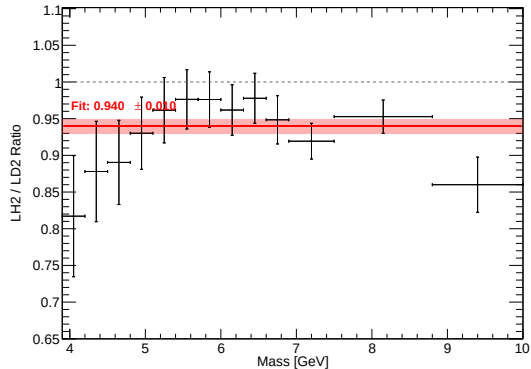


Overlay & Ratio (Mass sliced by p_T): p_T Bin 1

$0.32 \leq p_T < 0.49$

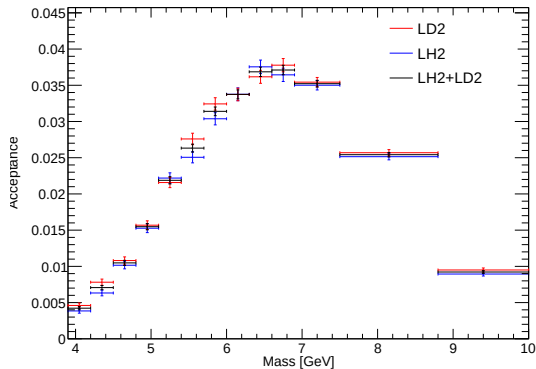


LH2/LD2 $0.32 \leq p_T < 0.49$

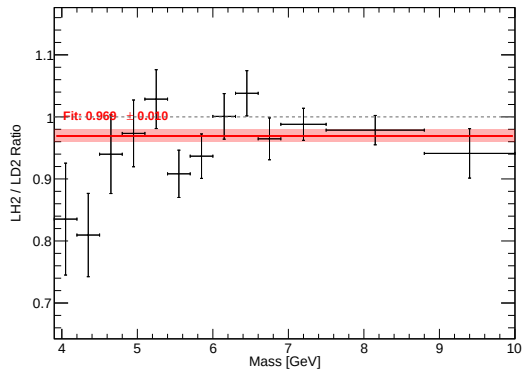


Overlay & Ratio (Mass sliced by p_T): p_T Bin 2

$0.49 \leq p_T < 0.63$

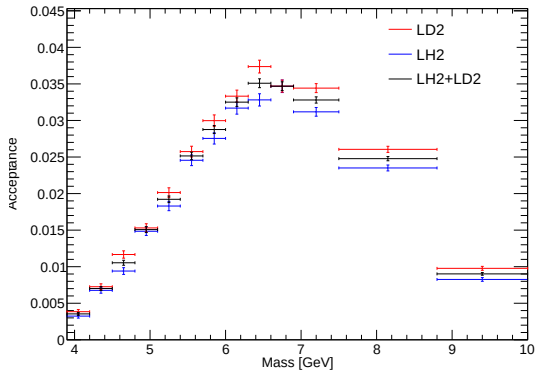


LH2/LD2 $0.49 \leq p_T < 0.63$

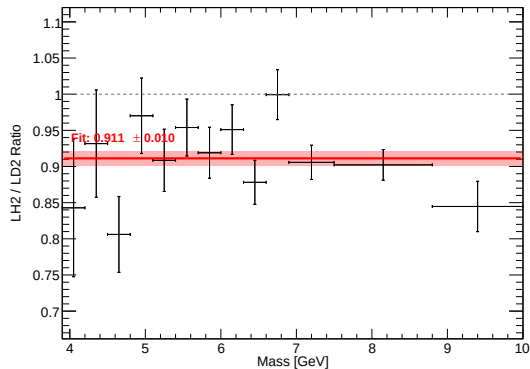


Overlay & Ratio (Mass sliced by p_T): p_T Bin 3

$0.63 \leq p_T < 0.77$

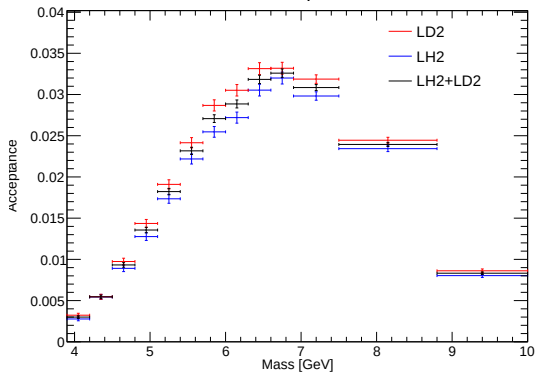


LH2/LD2 $0.63 \leq p_T < 0.77$

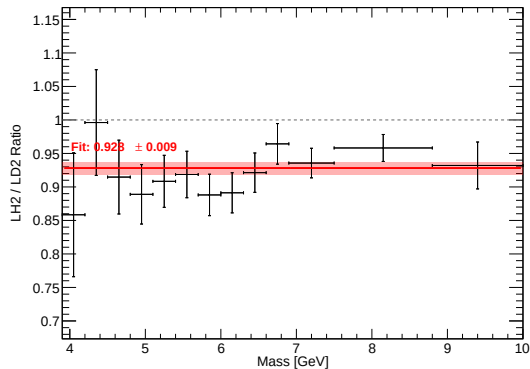


Overlay & Ratio (Mass sliced by p_T): p_T Bin 4

$0.77 \leq p_T < 0.95$

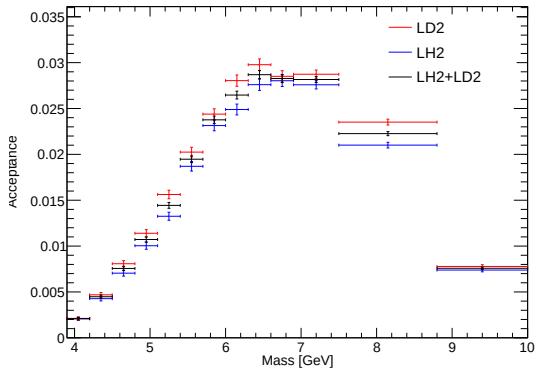


LH2/LD2 $0.77 \leq p_T < 0.95$

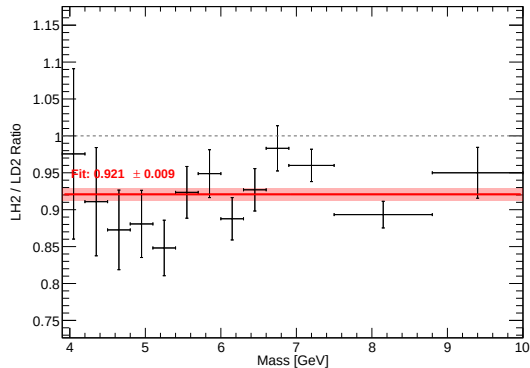


Overlay & Ratio (Mass sliced by p_T): p_T Bin 5

$0.95 \leq p_T < 1.18$

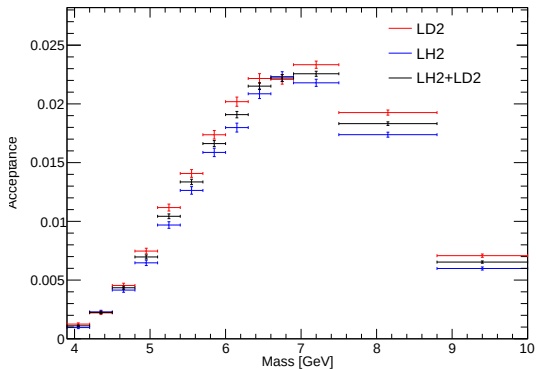


LH2/LD2 $0.95 \leq p_T < 1.18$



Overlay & Ratio (Mass sliced by p_T): p_T Bin 6

$1.18 \leq p_T < 1.80$



LH2/LD2 $1.18 \leq p_T < 1.80$

